

Anyonic Tensor Networks

Symmetries, Categories and Circuits

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INTRODUCTION

Tensor Networks allow the efficient representations and computation on quantum states for large N using methods such as DMRG. Symmetric Tensor Networks yield a speedup over traditional methods for appropriate Hamiltonians. Furthermore, we would also like to represent *Anyonic* states and Hamiltonians by these operations, and since some anyon species are universal for quantum computation, we would like to represent circuits as well. It is folklore that tensor networks and quantum systems in general are governed by category-theoretic constructions.

In this thesis we give a formal overview how all these concepts fit into the same categorical framework while presenting a review of the existing literature, i.e. we give an interpretation of the results of Singh and Vidal [SV12], Weichselbaum [Wei12b], Pfeifer et al. [Pfe+10], Unfried [Unf24], and Devos and Haegeman [DH25], embed them with a rigorous categorical theory and provide missing proofs. Due to the interconnectedness with topological order, we phrase them in the language of [KZ22].

1.1 THESIS OUTLINE

In Chapter 2 we review the technical literature which is shared by the chapters (sections?). We give accounts on symmetries, representation theory, fermionic networks, anyons and categories. Theorem 5.2.3 presents the main theorem that is the backend of our tensor system. While we will state the first 2 sections with scarce proofs, chapter 5 will be quite rigorous and supercede all statements from the previous sections.

Maybe we will present a outlook on Modular Categories

BACKGROUND

2.1 LITERATURE & RELATED WORK

Literature 2.1.1 (Tensor Networks). The modern theory of *tensor networks* shaped out of the study of *renormalization* and quantum many-body systems, initially with the *density matrix renormalization ansatz (DMRG)* of White [Whi92; Whi93] and later extended by Ostlund and Rommer [OR95] as a variational method (ansatz?) in the form of *matrix product states (MPSs)* (Maybe this is actually due to [FNW92]). This was further generalized to 2 dimensions with the introduction of *projected entangled-pair state (PEPS)* [VC04]. *multiscale entanglement renormalization ansatz (MERA)* [Vid08] can be considered as an extension of MPS allowing for long-range entanglement (this is probably wrong). Whats missing? Holography [SMB18] but there are earlier ones.

In the purely mathematical view as a network of tensor contractions however, both concept and the notation we use today were introduced by [Pen71] under the name of *tensor-systems*.

While we introduce this concept classically and endow it with symmetry as a start, we will start concerning ourselves with a categorical embedding of tensor networks in what is commonly referred to as a *fusion-category* (Lit ???).

Literature 2.1.2 (Symmetric Tensor Network). Originally, the idea to exploit symmetries of the hamiltonian for a speedup was proposed to be used in dmrg by [MG02] (What is [SM98]? This sounds like it as well.). More general theory for *globally invariant tensor networks*, i.e. networks that are built of tensors that are globally invariant is given by Singh, Pfeifer, and Vidal [SPV10] preceeding more detailed accounts on abelian symmetries [SPV11] and nonabelian (SU(2)) symmetries in [SV12]. A similar approach to (non)abelian symmetries is given by Weichselbaum [Wei12a; Wei20]. These rely on the general idea, that under representations of some compact group G , there are on site symmetries on each tensor, i.e. $[A, U_g^{\otimes k}] = 0$ there is a generalized *wigner-eckart-theorem* for tensors with n legs of the form

$$T \in \bigoplus_a D_a \otimes V_a$$

where D_a is a matrix and V_a is a *fusion-tree*, i.e. a clebsch-gordan tensor. Reviews and specific algorithms in the SU(2) case are given by [SO20; Sch20; Sch+20]. [Sch+20, § 8] proposes that this method is appropriate for Anyonic simulation as well (although this is done earlier by singh & pfeiffer.). Symmetries as categorified networks, as we will present in ??, are explained in [Unf24] and [DH25] and in this work in chapter 6.

For a list of software implementing symmetries see ??.

Literature 2.1.3 (Anyons). *Anyons* as quasiparticles with exotic exchange statistics were originally proposed by [Wil90] [LM77]. Anyons are usually described as *defects* in a (topologically ordered?) medium of dimension two (technically codimension after [SS23AnyonsInTedK]). The exchange statistics of N particles on this medium M is governed by the *configuration space*

$$\text{Conf}_N(M) = \{(x_1, \dots, x_N) \in M^N : x_i \neq x_j\} \setminus S_N.$$

The statistics of the particle exchange is given by the representation of $\pi_1(\text{Conf}_N(M))$ (the fundamental (homology?) group). If $M = \mathbb{R}^2$, this corresponds to the *braid group* on N strands B_N . If the dimension of

$M \geq 3$ this corresponds to the symmetric group S_N . If these representations are unitary and the (ground state?) is degenerate, this allows for *topological quantum computation* (cf. Literature 2.1.7).

Literature 2.1.4 (Anyonic Spin Chains and Tensor Networks). (Check if this is the first). A first appearance of *Anyonic Tensor Networks* in terms of spin chains is given by Feiguin et al. [Fei+07] as the *golden chain*, which realizes a interacting 2-local hamiltonian on a n Fib-anyon lattice. This is reviewed in [Tre+08] with generalizations to 3-local and longer range hamiltonians. A general account outside of the Fib-model can be found in [Gil+13] and [BG17] (Check these again.). We can consider these approaches as a sort of manualization of the tensor network approach. Anyonic tensor networks are first introduced by [Pfe+10] and [Pfe12], developed out of the symmetry methods from chapter 3 and both the *isotopy* rules and anyonic operators defined in [BSS08]. Furthermore, stuctures such as an anyonic MERA are introduced.

We also need to discuss the topic of *anyonic entanglement entropy*, as given in [Pfe14; BKP17; Pfe12].

Literature 2.1.5 (Simulation of Anyons and String Nets). This should probably go in the previous section. [Liu+22] [Kir+23]

Literature 2.1.6 (Anyons as Categories). The correspondence between the *physical* treatment of anyons and the *categorical treatment* of anyons as a *braided fusion category* (modular functor?) was originally descibed by Kitaev [Kit06, § E, § 8] based on previous work in quantum field theory already making use of modular categories and their diagrammatic calculus such as [RT91]. Further treatments are found in [wang] and [simon-nayak-das-sarma]. A modern treatment of the realization through the Configuration space of n -anyons if given by [90eb98357acd4634abd18d906cc3811d] and in terms of modular functors by [Bur16].

Literature 2.1.7 (Topological Quantum Computation). Fault tolerant *Topological Quantum Computation* (TQC) using anabelian anyons was originally proposed by Kitaev [Kit03, § 7] (Originally published in 1997). Also [Fre98] and ??? has shown that there are nonabelian anyon models that are universal [DN05] [Moc03] [Moc04].

[MSS24] gives nicer literature. [Sim23] is a great textbook account from one of the original authors on quantum compiling and computing. [Nay+08] is also one of the original accounts. This does not need to be any longer. Maybe mention categorical computation in general? but idk what htis is yet [KZ23].

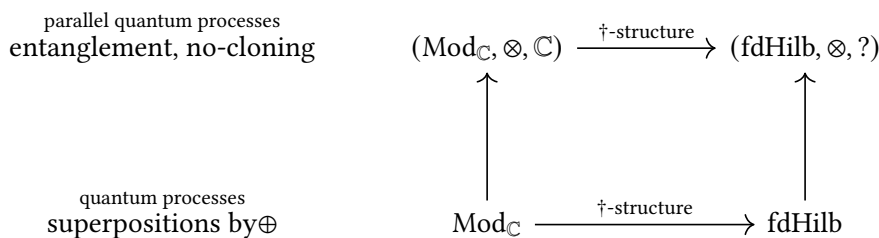
Literature 2.1.8 (Quantum Compiling). Some (Many?) anyon theories are universal for quantum computation. The *Solovay-Kitaev theorem* [Kit97][DN05] yields an efficient algorithm for approximating any quantum gate using only basic gates from a universal set. Here we can consider 1.) The representation of braid circuits as a tensor network operation and 2). A natural embedding of injection weaves as proposed by Simon et al. [Sim+06] where 2 anyons are braided in their composite fusion channel by realizing this operation as a tensor reshape.

Literature 2.1.9 (Categorified Quantum Mechanics). There are multiple approach to a diagramatic and categorified approach to quantum mechanics. Generally one wants both spaces and maps represented by a framework that allows for parallel and entangled operations, i.e. has a tensor product structure. [Sel11] gives a review of this [VW18] does for tensor networks specifically.

TODO: find the XZ-calculus literature by Kissinger? Specific examples such as XZ-calculus in the context of TQC is handled by [AK23].

TODO: How to mention that this results in or from Hilb_{fd} ? Does it? The work from selinger et.al does not yet include anyons. For this, one needs Literature 2.1.6.

Also, take [SS26, § 1] for a description of categorified mechanics. This stats that structure for *entangled* processes is given by the monoidal category of complex vector spaces.



The categorical language specifically yield interpretations of theorems such as no-cloning and no deleting [Abr12].

Additionally, many of the concepts of topological order are governed by category theory [KZ22], as well as the topic of higher symmetries (should this be its own Lit sect if relevant? It is for MPO symmetries [DT24] [Kon+20]).

2.2 ON TENSORS

Throughout, k denotes a field and all vector spaces are finite-dimensional over k .

A *tensor* is an element of a tensor product of finite-dimensional vector spaces,

$$T \in \bigotimes_{j \in J} V_j,$$

and we call T a rank $|J|$ tensor. The *dimension* of the tensor is then $\dim(T) = \prod_{j \in J} \dim(V_j)$.

The *dual space* of a vector space V is $V^* = \text{Hom}(V, k)$, the space of linear functionals $f: V \rightarrow k, x \mapsto f(x)$. In Dirac notation we write $|v\rangle \in V$ for vectors (kets) and $\langle f| \in V^*$ for linear functionals (bras), so that $\langle f|v\rangle = f(v)$.

Since all spaces are finite-dimensional, we have the canonical isomorphisms

$$V^{**} \cong V, \quad (V \otimes W)^* \cong V^* \otimes W^*, \quad W \otimes V^* \cong \text{Hom}(V, W),$$

where the last is given on simple tensors by

$$|w\rangle \otimes \langle f| \mapsto (v \mapsto \langle f|v\rangle |w\rangle),$$

i.e. $|w\rangle\langle f|$ is the outer product.

Because $V^{**} \cong V$, any tensor $T \in V_1 \otimes \cdots \otimes V_n$ can be regarded as a linear map after choosing a split of its legs at some $0 \leq m \leq n$:

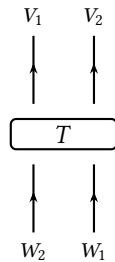
$$T: V_1^* \otimes \cdots \otimes V_m^* \longrightarrow V_{m+1} \otimes \cdots \otimes V_n.$$

Example 2.2.1. Let $t \in V_1 \otimes V_2 \otimes W_2^* \otimes W_1^*$. Choosing the split after V_2 , t is an operator $W_1 \otimes W_2 \rightarrow V_1 \otimes V_2$, which in Dirac notation reads

$$t = \sum_{i_1, i_2, j_1, j_2} T_{j_1 j_2}^{i_1 i_2} |v_{i_1}^1\rangle |v_{i_2}^2\rangle \langle w_{j_2}^2| \langle w_{j_1}^1|,$$

where $\{|v_i^a\rangle\}$ is a basis of V_a and $\{\langle w_j^b|\}$ is a basis of W_b^* (the dual basis of a basis of W_b).

Graphical Notation 2.2.2. Graphically, a tensor $T: W_n \otimes \cdots \otimes W_1 \rightarrow V_1 \otimes \cdots \otimes V_m$ is a coupon with a leg for each factor of its spaces: legs for the codomain are outgoing from the top of the coupon, legs for the domain point inwards towards the bottom of the diagram. Arrows are oriented so that a leg carrying V points up and



a leg carrying V^* points down.

Two tensors with matching legs are contracted using the *evaluation* and *coevaluation* maps

$$\begin{aligned} \text{ev}_V: V^* \otimes V &\rightarrow k, & f \otimes v &\mapsto f(v), \\ \text{coev}_V: k &\rightarrow V \otimes V^*, & \lambda &\mapsto \lambda \sum_i v_i \otimes f_i, \end{aligned}$$

where $\{v_i\}$ is a basis of V and $\{f_i\}$ its dual basis. A *contraction* is a composite of identities and evaluation maps that removes the paired legs from the tensor product. For example, for operators $A \in U \otimes V^*$ (a map $V \rightarrow U$) and $B \in V \otimes W^*$ (a map $W \rightarrow V$), the product $AB \in U \otimes W^*$ is

$$AB = (\text{id}_U \otimes \text{ev}_V \otimes \text{id}_{W^*})(A \otimes B).$$

Equip U, V, W with bases $\{u_a\}, \{v_i\}$ and w_b respectively. We can now write

$$A \otimes B = \sum_{a,i,j,b} A_{ai} B_{jb} |u_a\rangle \otimes \langle v_i| \otimes |v_i\rangle \otimes \langle w_b|$$

and by evaluation $\text{ev}_V(\langle v_j| \otimes |v_i\rangle) = \langle v_j|v_i\rangle = \delta_j^i$ we can obtain

$$AB = \sum_{a,i,b} A_{ai} B_{ib} |u_a\rangle \otimes \langle w_b|$$

with coordinates $(AB)_b^a = \sum_i A_i^a B_b^i$ which is precisely the well known matrix-multiplication formula.

Remark 2.2.3. Here we have chosen the vector spaces V_j to be abstract vector spaces, i.e. they need not be of the form k^d , and all the statements like duality and contraction are generally basis-independent. Making a choice of basis identifies $V \cong k^{\dim V}$ and turns T into a coordinatized array as in the example above.

Traditionally, for tensor networks we care about local Hilbert space, so our factors are just $V_j = \mathbb{C}^{d_j}$. Tensors are thus multidimensional arrays and contractions are summations over a shared index.

Later on, we will see version of tensor networks that do not have a basis of coordinatized vectors, but rather a abstract indexable basis of formal symbols.

It should also be mentioned that on the implementation level, we also have to consider how we store these tensors. If we have real tensors we may use datatypes such as `float64`, if we have complex tensors we can use `complex128`. Generally, we store these tensors as arrays (vectors?) in contiguous memory.

If we take the definition of a tensor over a field such as $\mathbb{C}$, we can store this on a computer in contiguous memory. Furthermore, if we have a vector in $\mathbb{C}^{d^2}$, we can equivalently choose to view this as a matrix in $\mathbb{C}^{d \times d}$. This however is not a basis independent process. If we view this vector instead as a member of the hom space $\text{Hom}(\mathbb{C}, \mathbb{C}^{d^2})$.

2.2.1 Tensor Networks & Matrix Product States

Often times we will consider quantum state on a lattice. For 1-dimensional systems, we can consider a lattice of n sites where each holds a local site $|i\rangle$. We can give an arbitrary state on this lattice as

$$|\Psi\rangle = \sum_{i_1 \dots i_n} c_{i_1 \dots i_n} |i_1 i_2 \dots i_n\rangle.$$

Storing this state requires representing the exponentially many coefficient $c_{i_1 \dots i_n}$. As an example, we may have two sites of spin 1/2, so we have a total state space of dimension 4 spanned by $|i_1 i_2\rangle \cong \mathbb{C}^4$ with $i_1, i_2 \in \{\uparrow, \downarrow\}$. Then, we can give the coefficients to obtain the state

$$\Psi^+ = \frac{1}{\sqrt{2}} (|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle)$$

and we can store this state as a 4×1 matrix (a statevector of dimension 4). If we keep the spin basis on the lattice sites, a matrix on n sites is thus representable by a statevector $|\Psi\rangle \in \mathbb{C}^{2^n}$.

Now, we are allowed to apply a reshape operation and change $\mathbb{C}^{2^n}$ into $\mathbb{C}^{2 \times 2^{n-1}}$ which corresponds to a tensor of order 2, i.e. a matrix with 2 columns and 2^{n-1} rows. Now that we have a matrix, we can apply the **singular value decomposition (SVD)** yielding a decomposition

$$c_{i_1, (i_2 \dots i_n)} = \sum_{j_1} U_{i_1, j_1} S_{j_1, j_1} (V^\dagger)_{j_1, (i_2 \dots i_n)}$$

Definition 2.2.4. A **Matrix Product State** is a quantum state $|\psi\rangle$ with n legs written as the contraction of n trivalent tensors $\{A^{[i]}\}_{i \in [n]}$. Formally this reads

$$|\psi[A]\rangle = \sum_{i_1, \dots, i_n} \text{Tr}(A^{i_1} A^{i_2} \dots A^{i_n}) |i_1 i_2 \dots i_n\rangle,$$

where each A is of the form

$$A^{[i]} = \sum_{i\delta_i \delta_{i+1}} A_{\delta_i \delta_{i+1}}^i |i\rangle \otimes \langle \delta_i| \otimes \langle \delta_{i+1}|$$

where $A_{ik}^{[i]} \in \mathbb{C}^{D_i \times D_{i+1}}$. The elements of the collection $\{D_i\}_{i \in [n]}$ is referred to as the **bond-dimensions**.

rephrase!

Link, rephrase.

Here? or where?

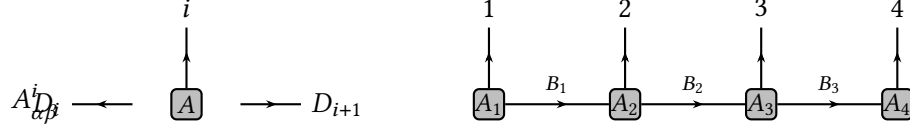
Is this correct for the storage?

Technically this is all redundant, as the indices are themselves specified by n other indices.

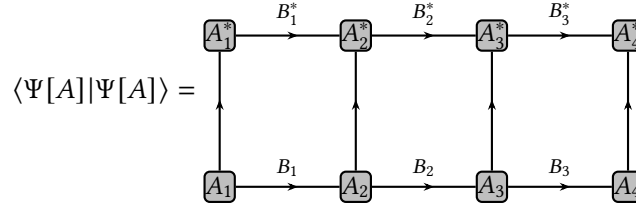
Note that this is a state with closed boundaries. If we want open boundaries we have to replace the trace with appropriate end vectors (nice source?).

Example 2.2.5. We can represent each product state $|\Psi\rangle = |\psi\rangle^{\otimes n}$ as a MPS with bond dimension 1.

Graphical Notation 2.2.6. We represent a MPS tensor to the left and a full MPS with $N = 4$ to the right.



Note that we can easily define an inner product on an MPS (as it represents a state) and we can view this as



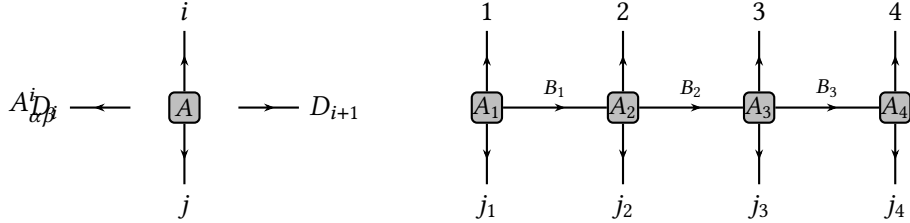
Definition 2.2.7. A *matrix product operator (MPO)* is an operator on a state with n legs on each side written as the contraction of 4 legged tensors $W[i], \dots$

$$\sum_{i_1, \dots, i_n, j_1, \dots, j_n} \text{tr}(M^{i_1 j_1} M^{i_2 j_2} \dots M^{i_n j_n}) |i_1 \dots i_n\rangle \langle j_1 \dots j_n|$$

We again use the convention of denoting the bonds by D , and we can regard each mps tensor as a mapping

$$\text{Hom}(P_i \otimes D_i, P_j \otimes D_{i+1})$$

Graphical Notation 2.2.8. We represent a MPO tensor to the left and a full MPO with $N = 4$ to the right.

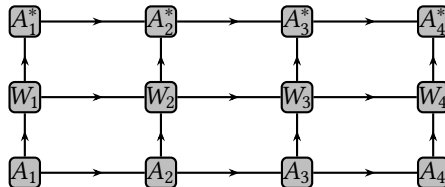


Constructing these operators can be done by the use of a *finite state machine* [CB08]. As an example, the XXZ-Hamiltonian is represented by the MPO tensors

$$W_{ij} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ S^x & 0 & 0 & 0 & 0 \\ S^y & 0 & 0 & 0 & 0 \\ S^z & 0 & 0 & 0 & 0 \\ 0 & S^x & S^y & S^z & 1 \end{pmatrix}_{ij} \quad (2.1)$$

with bond dimension 5, note that this acts on the physical states as such.

As an application, we can immediately represent the expectation value of a state $\langle \Psi | H | \Psi \rangle$ by using MPSs and MPOs:



SYMMETRIC TENSOR NETWORKS

This section develops the theory underlying *symmetric tensor networks*, specifically for non-abelian groups. We give an informal overview on the necessary representation theoretic concepts in Section 3.1. We employ the example of the XXZ-Hamiltonian for both abelian and non-abelian symmetries and give an exact calculation of its conserved values. Here we are mostly interested in the *speedup* obtained by exploiting the emergent block-structure of symmetric operators. Most of the proofs and formal definitions required for this formalism are delegated to Chapter 5, however there will still be repetitions of some of the theory.

3.1 REPRESENTATION THEORY

This section gives a quick review of representation theory. We consider all vector spaces and their representations here to be finite, the groups may only be compact. For some of the more basic definitions as those of a group and some of the matrix groups we are using here, see Appendix A.1.

Definition 3.1.1 (Representation). A *representation* of a compact group G over a vector space V is a group homomorphism

$$\rho : G \rightarrow \text{GL}(V), \quad \rho(g)\rho(h) = \rho(g \cdot h), \quad \rho(e) = \text{id}_V$$

The *representation space* is denoted by the tuple (ρ, V) but usually we shorten this to just V .

Equivalently we can state the definition of a representation of G as imposed by its action on V , given as a map

$$\alpha : G \times V \rightarrow V, \quad (g, v) \mapsto g \cdot v \tag{3.1}$$

such that this is linear, todo. I think we get the equality to the hom def if we set $g \cdot v = \rho(g)v$

Definition 3.1.2. A linear map $A : V \rightarrow W$ is an *intertwiner*, if under representations $\rho_V : G \rightarrow \text{GL}(V)$ and $\rho_W : G \rightarrow \text{GL}(W)$ it holds that

$$A \circ \rho_V(g) = \rho_W(g) \circ A \quad \forall g \in G.$$

Denote the space of all intertwiners between V and W of the group G as $\text{Hom}_G(V, W)$, similarly denote its endomorphisms as $\text{End}_G(V) := \text{Hom}_G(V, V)$.

Definition 3.1.3 (Unitary representation). If our vector space V is equipped with a (Hermitian?) inner product

Definition 3.1.4 (Irreducible representation (Woit 16)). A representation ρ is an *irrep* if there are no $W \subset V$ such that $(\rho|_W, W)$ is a representation.

Definition 3.1.5. Given reps (ρ_V, V) and (ρ_W, W) , the *tensor product representation* $(\rho_{V \otimes W}, V \otimes W)$ is defined by its action on the elements

$$\rho_{V \otimes W}(g)(v \otimes w) = \rho_V(g)v \otimes \rho_W(g)w$$

Definition 3.1.6. As above, the *direct sum representation* $(\rho_{V \oplus W}, V \oplus W)$

$$\rho_{V \oplus W}(g) = \begin{pmatrix} \rho_V(g) & \\ & \rho_W(g) \end{pmatrix}$$

cite rep theory textbook.

This is quite important.

Definition 3.1.7 (Isotypic / Semisimple decomposition). Each semisimple representation V decomposes as

$$V \cong \bigoplus_{i \in I} V_i^{\oplus n_i} \cong \bigoplus_{i \in I} \mathbb{C}^{n_i} \otimes V_i \cong \bigoplus_{i \in I} V_i \otimes \mathbb{C}^{n_i}$$

by grouping isomorphic simples. Is n_i unique?

Due to a theorem by Peter & Weyl [**<empty citation>**], the unitary representations of any compact G are semisimple.

Therefore, it must hold for any irreps R_a, R_b of G that their tensor product is completely reducible with the decomposition

$$R_a \otimes R_b \cong \bigoplus_{c \in I} R_c^{\oplus N_{ab}^c},$$

with N_{ab}^c called *fusion-multiplicities*. If

$$N_{ab}^c \leq 1 \quad \forall a, b, c$$

the decomposition is said to be *multiplicity-free*.

Note as well that the appearance of the *direct sum* here (cf. Equation (5.7)) is fully characterized by its injection and projection maps, for which we provide a graphical calculus.

$$|a, b; c, \mu\rangle : R_c \rightarrow R_a \otimes R_b \quad \text{and} \quad \langle c, \mu; a, b| : R_a \otimes R_b \rightarrow R_c$$

$$\begin{array}{ccc} \begin{array}{c} R_c \\ \downarrow \\ \mu \\ \swarrow \quad \searrow \\ R_a \quad R_b \end{array} & & \begin{array}{c} R_a \quad R_b \\ \swarrow \quad \searrow \\ \mu \\ \downarrow \\ R_c \end{array} \end{array} \quad (3.2)$$

These maps are more commonly known as *Clebsch-Gordan tensors*. Note that the equations

$$\langle \nu, d; a, b| |a, b; c, \mu\rangle = \delta_{\mu\nu} \delta_{cd} \text{id}_{R_c} \quad (3.3)$$

and

$$\sum_{c, \mu} |a, b; c, \mu\rangle \langle \mu, c; a, b| = \text{id}_{R_a \otimes R_b}. \quad (3.4)$$

hold. As a consequence of this relation, the outer product of two Clebsch-Gordan spans the space $\text{End}_G(R_a \otimes R_b)$.

On the operators between tensor products, i.e. $T \in \text{End}_G(R_a \otimes R_b)$ this yields a formula

$$T \cong \bigoplus_c \text{id}_{R_c} \otimes \mathbb{C}^{N_{ab}^c}$$

Example 3.1.8. The simple objects of $G = \text{U}(1)$ are given by the one-dimensional representations

$$\chi_n : \text{U}(1) \rightarrow \mathbb{C}^\times, \quad e^{i\theta} \mapsto e^{in\theta}$$

and each representation of $\text{U}(1)$ decomposes as

$$V \cong \bigoplus_{n \in \mathbb{Z}} \chi_n^{\oplus m_n}.$$

Its fusion rule is given by

$$\chi_n \otimes \chi_m = \chi_{n+m}. \quad (3.5)$$

Example 3.1.9. Let $G = \text{SU}(2)$ and let V_i and V_j be two spin irreps of charge i and j . These combine to

$$(\pi_{V_i \otimes V_j}, V_i \otimes V_j) = \bigoplus_{k=|i-j|}^{i+j} (\pi_{V_k}, V_k) \quad (3.6)$$

in what is referred to as the *Clebsch-Gordan-series*.

Example 3.1.10. Lets look at an example of spin irreps in the $SU(2)$ theory. The combined basis of two spin- $\frac{1}{2}$ spaces is given by the collection $\{|\uparrow\rangle, |\downarrow\rangle\}^{\otimes 2}$. By the fusion rule eq. (3.6), the spins combine as

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

with its associated basis $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle, |0, 0\rangle\}$. We call this basis the *coupled basis*, indexed as $|j, j_z\rangle$ where j is the total spin and j_z is the magnetization? However we can also give a formal notation for this. Denote a state in the spin basis as $|j, m\rangle$ where j is the spin and m is its magnetization? In the case that we have singular spin $1/2$ states, the second labels is redundant since

$$|\uparrow\rangle = \left| \frac{1}{2}, \frac{1}{2} \right\rangle \quad |\downarrow\rangle = \left| \frac{1}{2}, -\frac{1}{2} \right\rangle.$$

Read up on this here.

Hmm i dont like this being an example. Should we even use the Lie theory here? I dont think we can get around it so put the intro in the appendix?

3.2 APPLICATION: THE XXZ HAMILTONIAN

The motivation for constructing symmetric tensor networks is as follows: Assume we have a (unitary) representation of a group G denoted $U(g)$. Furthermore assume we have a Hamiltonian that commutes with $[U(g)^{\otimes L}, H] = 0$, equivalently, H omits a *global symmetry* $U(g)^{\otimes L} H (U(g)^{\otimes L})^\dagger$. Then there are different presentations of H with respect to a certain basis by *simultaneous diagonalization* [HJ12, Thm 1.3.12], as shown in Figure 3.1. Notably these presentation induce a *block-sparse* structure on the respective operators.

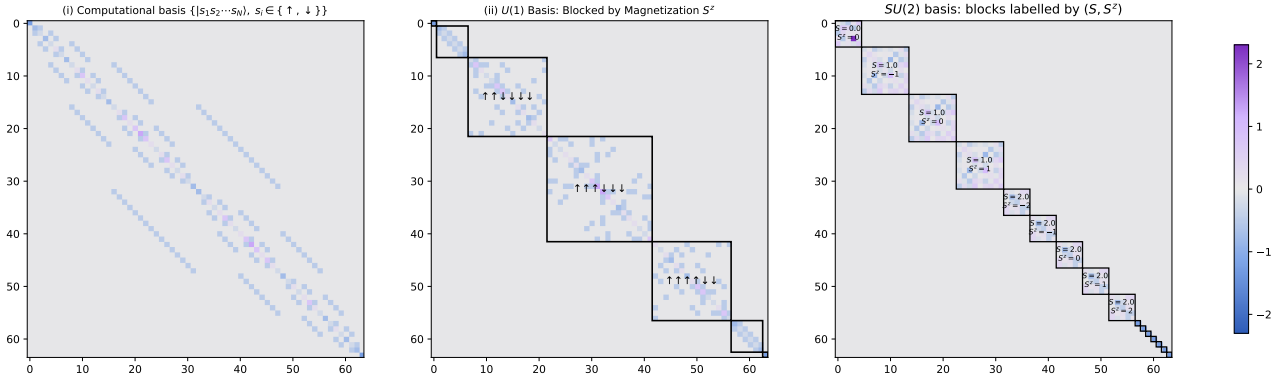


Figure 3.1: Three different presentations of the XXZ Hamiltonian for $N = 6$. (i): In the default computational basis of spin vectors. (ii) In a reordering of the computational basis into equal spin up (down) classes, i.e. no spin up, one spin up etc..., here U is simply a permutation matrix. (iii) After changing the basis into the (S^2, S^z) basis.

We now show how to obtain this block-sparse structure and more by example of the XXZ hamiltonian:

Definition 3.2.1. The *XXZ Hamiltonian* on N sites is

$$H = J \cdot \sum_{i=1}^{N-1} (S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta S_i^z S_{i+1}^z), \quad (3.7)$$

acting on the basis $|s_1 s_2 \dots s_n\rangle$ where $s_i \in \{\uparrow, \downarrow\}$.

Note that by definition, the XXZ-Hamiltonian is a local hamiltonian of the form

$$H = \sum_i h_{i,i+1}$$

This operator admits both a $U(1)$ symmetry, and a $SU(2)$ symmetry for $\Delta = 1$. We will handle the (simpler) abelian case first. The representations of $U(1)$ here are induced by the operator

$$S_{\text{tot}}^z = \sum_{i=1}^N S_i^z \quad \text{with action} \quad U(\theta) : \theta \in U(1) \mapsto e^{-i\theta 2S_{\text{tot}}^z}$$

It can be shown that H is invariant under $U(\theta)$, and it suffices to show this invariance for the generators S_{tot}^z :

$$U(\theta)HU(\theta)^\dagger = H \quad \Longleftrightarrow \quad [H, S_{tot}^z] = 0. \quad (3.8)$$

suffices by A.3

Diagonalizing both operators yields a block structure as in Figure 3.1(ii). Note however that the diagonalization here essentially produced a permutation matrix, and we just reordered the basis to group by total spins. If we want to state this in terms of irreps, we have to realize that $\mathbb{C}^2 \cong \chi_{-1} \oplus \chi_1$. By eq. (3.5) we can state this space as the direct sum

$$\begin{aligned} (\mathbb{C}^2)^{\otimes 6} &\cong \chi_{-6} \oplus \chi_{-4}^{\oplus 6} \oplus \chi_{-2}^{\oplus 15} \oplus \chi_0^{\oplus 20} \oplus \chi_2^{\oplus 15} \oplus \chi_4^{\oplus 6} \oplus \chi_6 \\ &\cong (\mathbb{C}^1 \otimes \chi_{-6}) \oplus (\mathbb{C}^6 \otimes \chi_{-4}) \oplus (\mathbb{C}^{15} \otimes \chi_{-2}) \oplus (\mathbb{C}^{20} \otimes \chi_0) \oplus (\mathbb{C}^{15} \otimes \chi_2) \oplus (\mathbb{C}^6 \otimes \chi_4) \oplus (\mathbb{C}^1 \otimes \chi_6). \end{aligned}$$

Recall that for $U(1)$, all representations are 1-dimensional, then by Schur's lemma, we can write H as its operator in $\text{End}_{U(1)}((\mathbb{C}^2)^{\otimes 6})$ as

$$H \cong \bigoplus_{S \in \{-6, -4, -2, 0, 2, 4, 6\}} \text{Mat}_{m_S \times m_S}(\mathbb{R}) \otimes \text{id}_{\chi_S} \cong \bigoplus_{S \in \{-6, -4, -2, 0, 2, 4, 6\}} \text{Mat}_{m_S \times m_S}(\mathbb{R}) \quad \text{where } m_S \in \{1, 6, 15, 20, 15, 6, 1\}$$

and so we should expect 7 blocks in total, just as given in Figure 3.1. To store these blocks, we need to store an effective $1^2 + 6^2 + 15^2 + 20^2 + 15^2 + 6^2 + 1^1 = 934$ floating point numbers.

Now lets consider the $SU(2)$ case, setting $\Delta = 1$. Here, the generators of $SU(2)$ are given by

$$S_{tot}^\alpha = \sum_{i=1}^N S_i^\alpha, \quad \alpha \in \{x, y, z\},$$

and its representation is $U(R) = e^{i\theta n \cdot S_{tot}}$. The Hamiltonian is invariant under

$$U(R)HU(R)^\dagger = H, \quad R \in SU(2)$$

Since

$$[S_{tot}^x, S_{tot}^y] = iS_{tot}^z$$

we need a new operator for the simultaneous diagonalization

$$S_{tot}^2 = \sum_{a \in \{x, y, z\}} (S_{tot}^a)^2$$

and we need to verify this is indeed the correct operator TODO.

The eigenvalues of S_{tot}^2 are $S(S+1)$ where S are the half integers.

Again, we can solve this by diagonalizing all 3 operators? Do this calculation as in the .ipynb and put it in the appendix.

Or we can again do the representation theoretical approach. Let V_j denote the spin- j irrep of $SU(2)$. We have that $\mathbb{C}^2 \cong V_{\frac{1}{2}}$ and by ?? that

$$(\mathbb{C}^2)^{\otimes 6} \cong V_{\frac{1}{2}}^{\otimes 6} \cong 5V_0 \oplus 9V_1 \oplus 5V_2 \oplus V_3 \cong (\mathbb{C}^5 \otimes V_0) \oplus (\mathbb{C}^9 \otimes V_1) \oplus (\mathbb{C}^5 \otimes V_2) \oplus (\mathbb{C} \otimes V_3)$$

Now our operator is a morphism in $\text{End}_G(V_{\frac{1}{2}}^{\otimes 6})$ and thus by Schur's lemma

$$H \cong \bigoplus_{S \in \{0, 1, 2, 3\}} \text{Mat}_{m_S \times m_S}(\mathbb{C}) \otimes I_{2S+1} \cong \bigoplus_{S \in \{0, 1, 2, 3\}} I_{2S+1} \otimes \text{Mat}_{m_S \times m_S}(\mathbb{C}). \quad (3.9)$$

So while we only have to store the matrix elements as blocks, i.e. here we have to store only four of them, in our picture we get 16 blocks because of the multiplicities. If we were to store this Hamiltonian as its initial matrix, we would have to store $64 \times 64 = 4096$ floats. If we only store the 4 blocks, this is reduced to $5^2 + 9^2 + 5^2 + 1^2 = 132$ floats¹. If we take the case that N is even larger, i.e. the case $N = 10$, the full operator

So this is quite strange, i have to use $2S_{tot}^z$ here, otherwise we get noninteger weights which makes no sense.

requires 1 048 576 floats while the symmetric operator only needs 16 796. Generally we have about two orders of magnitude less.

While this big picture overview conveys the general idea, we should do some of the calculations by hand. In order to get an interpretation of the matrix coefficients in eq. (3.9), we restrict ourselves to the case of the XXZ-Hamiltonian for $N = 2$ spins:

$$H = J \cdot (S_1^x S_2^x + S_1^y S_2^y + S_1^z S_2^z) = J \cdot S_1 \cdot S_2$$

This acts on the space $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ and thus on the basis $\{|\uparrow\rangle, |\downarrow\rangle\}^{\otimes 2}$. This has two distinct fusion outcomes, V_0 and V_1 . The basis of the coupled states is given by $|1, 1\rangle, |1, 0\rangle, |1, -1\rangle$ (the triplet state) and $|0, 0\rangle$ (the singlet state), and we want to reformulate $S_1 \cdot S_2$ in the coupled spin basis. We can do this purely algebraically by rearranging

$$S^2 = S_1^2 + S_2^2 + 2(S_1 \cdot S_2) \implies S_1 \cdot S_2 = \frac{1}{2}(S^2 - S_1^2 - S_2^2).$$

Since $S_i^2 = s(s+1)$ is fixed for $s = \frac{1}{2}$ here, we ought to write

$$H = \frac{J}{2}(S^2 - 3/2)$$

and this again splits into the sectors for $S = 0$ and $S = 1$

$$c_0 = \frac{J}{2}(0 - 3/2) = -\frac{3}{4}J \quad \text{and} \quad c_1 = \frac{J}{2}(2 - 3/2) = \frac{1}{4}J$$

which are exactly the two degrees of freedom we get by the direct sum. So we expect our operator, now in

$\bigoplus_{S \in \{0,1\}} \text{End}_G(V_S)$ to look like

$$\left(\begin{array}{c|ccc} -\frac{3}{4}J & 0 & 0 & 0 \\ \hline 0 & \frac{1}{4}J & 0 & 0 \\ 0 & 0 & \frac{1}{4}J & 0 \\ 0 & 0 & 0 & \frac{1}{4}J \end{array} \right).$$

However we can also still represent this as an element of $\text{End}_G(V_{1/2}^{\otimes 2})$ *factored* through its fusion result. This notation might for now come out of nowhere, but remember how we characterized the graphical calculus for the Clebsch-Gordan tensors in eq. (3.2). The graphical calculus here reads: Change base into the coupled basis, apply the two degrees of freedom that we just calculated, and change the base back into the uncoupled basis.

$$-J \frac{3}{4} \cdot \begin{array}{c} \frac{1}{2} \quad \frac{1}{2} \\ \searrow \quad \swarrow \\ \downarrow 0 \\ \swarrow \quad \searrow \\ \frac{1}{2} \quad \frac{1}{2} \end{array} + \frac{J}{4} \cdot \begin{array}{c} \frac{1}{2} \quad \frac{1}{2} \\ \searrow \quad \swarrow \\ \downarrow 1 \\ \swarrow \quad \searrow \\ \frac{1}{2} \quad \frac{1}{2} \end{array} \quad (3.10)$$

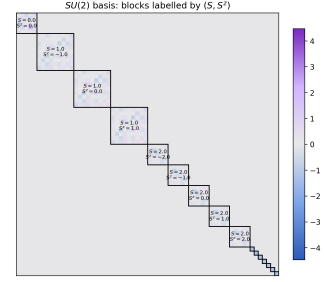
Now we consider the same Hamiltonian with $N = 3$, our state space is $V_{1/2}^{\otimes 3} = V_{3/2} \oplus V_{1/2} \oplus V_{1/2}$ with a multiplicity 2 on the $1/2$ state and the Operator is

$$H = \frac{1}{2}(S_1 \cdot S_2 + S_2 \cdot S_3)$$

and consider that we chose the left bracketing when combining the spins. Note that this means that the $S_1 \cdot S_2$ operator is fully diagonal, but the $S_2 \cdot S_3$ operator has off diagonal interactions. The diagonalized Hamiltonian is given on the right.

¹As we will see when discussing the actual algorithms behind these symmetric tensors, there is still more structure to be stored. However, these methods will still be more efficient than storing the full numerical Hamiltonian. We do exact calculations in chapter 5.

$$S_2 \cdot S_3 = \underbrace{\text{id}_{V_{1/2}}}_{\dim 2} \otimes \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & 0 \end{pmatrix} \oplus \underbrace{\text{id}_{V_{3/2}}}_{\dim 4} \otimes (1/4)$$



Visually we can define the operators before and include the ground state in a tree-like fashion:

$$\begin{aligned}
& -J\frac{3}{4} \otimes 0 \quad + \quad -J\frac{3}{4} \otimes 1 \quad + \\
& \begin{array}{c} \frac{1}{2} \quad \frac{1}{2} \\ \swarrow \quad \searrow \\ \text{---} \uparrow \text{---} \\ \swarrow \quad \searrow \\ \frac{1}{2} \quad \frac{1}{2} \\ \swarrow \quad \searrow \\ 0 \quad \nearrow \\ \swarrow \quad \searrow \\ \text{---} \uparrow \text{---} \end{array} \quad \begin{array}{c} \frac{1}{2} \quad \frac{1}{2} \\ \swarrow \quad \searrow \\ \text{---} \uparrow \text{---} \\ \swarrow \quad \searrow \\ \frac{1}{2} \quad \frac{1}{2} \\ \swarrow \quad \searrow \\ 0 \quad \nearrow \\ \swarrow \quad \searrow \\ \text{---} \uparrow \text{---} \end{array} \\
& \frac{\sqrt{3}}{4} \otimes \begin{array}{c} 1 \quad 2 \quad 3 \\ \swarrow \quad \searrow \quad \nearrow \\ \text{---} \uparrow \text{---} \\ \swarrow \quad \searrow \quad \nearrow \\ 1 \quad 2 \quad 3 \end{array} \quad \frac{\sqrt{3}}{4} \otimes \begin{array}{c} 1 \quad 2 \quad 3 \\ \swarrow \quad \searrow \quad \nearrow \\ \text{---} \uparrow \text{---} \\ \swarrow \quad \searrow \quad \nearrow \\ 1 \quad 2 \quad 3 \end{array}
\end{aligned}
\tag{3.11}$$

3.3 SYMMETRIC TENSORS

So far we have applied the symmetric formalism to Hamiltonians, which are endomorphisms $\text{End}_G()$ between some finite vector spaces, and we have an interpretation of local Hamiltonians acting between sites. This section will show how to express arbitrary symmetric tensors as intertwiners, and thus also constructing symmetric MPSs, MPOs and MERAs.

Let G be a group, and all representations thereof are finite-dimensional.² Fix representations $\rho_i: G \rightarrow \mathrm{GL}(V_i)$ for $i = 1, \dots, n$ and $\sigma_j: G \rightarrow \mathrm{GL}(W_j)$ for $j = 1, \dots, m$, and set

$$\begin{aligned} V &= V_1 \otimes \cdots \otimes V_n, & \rho_V(g) &= \rho_1(g) \otimes \cdots \otimes \rho_n(g), \\ W &= W_1 \otimes \cdots \otimes W_m, & \rho_W(q) &= \sigma_1(q) \otimes \cdots \otimes \sigma_m(q). \end{aligned}$$

The dual representation on W_j^* is $\sigma_j^*(g)f = f \circ \sigma_j(g)^{-1}$. A tensor with n outgoing and m incoming legs is an element

$$T \in V_1 \otimes \cdots \otimes V_n \otimes W_m^* \otimes \cdots \otimes W_1^* \cong V \otimes W^*,$$

on which G acts by

$$\Pi(q) = \rho_1(q) \otimes \cdots \otimes \rho_n(q) \otimes \sigma_m^*(q) \otimes \cdots \otimes \sigma_1^*(q). \quad (3.12)$$

(The reversed order on the dual legs is the nested convention for $(W_1 \otimes \cdots \otimes W_m)^*$; since the factors act on different legs, it only permutes tensor factors.)

²Compactness of G is not needed here; but we require it later for Schur's lemma. Or where?

Definition 3.3.1. A tensor $T \in V \otimes W^*$ is *symmetric* (or G -invariant) if $\Pi(g)T = T$ for all $g \in G$ [SPV10].

Let $\Phi: V \otimes W^* \rightarrow \text{Hom}(W, V)$ be the linear isomorphism $\Phi(v \otimes f) = (w \mapsto f(w)v)$.

Lemma 3.3.2. For all $g \in G$ and $T \in V \otimes W^*$,

$$\Phi(\Pi(g)T) = \rho_V(g) \circ \Phi(T) \circ \rho_W(g)^{-1}.$$

Consequently T is symmetric if and only if $\Phi(T) \in \text{Hom}_G(W, V)$, i.e.

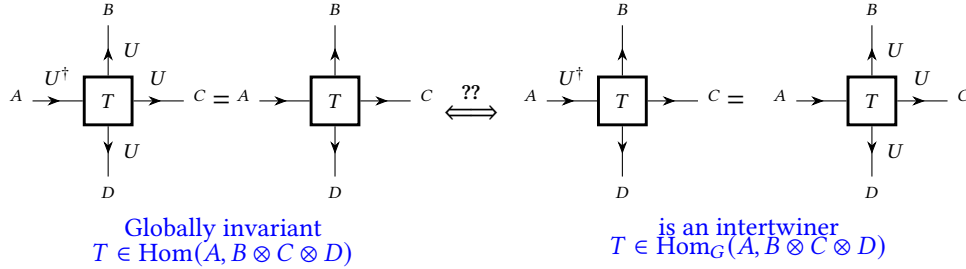
$$\rho_V(g) \circ \Phi(T) = \Phi(T) \circ \rho_W(g) \quad \forall g \in G.$$

Proof. Both sides are linear in T , so take $T = v \otimes f$. Then $\Pi(g)T = \rho_V(g)v \otimes (f \circ \rho_W(g)^{-1})$, and

$$\Phi(\Pi(g)T)(w) = f(\rho_W(g)^{-1}w) \rho_V(g)v = \rho_V(g) \Phi(T)(\rho_W(g)^{-1}w),$$

which is the first claim. Since Φ is injective, $\Pi(g)T = T$ holds iff $\rho_V(g)\Phi(T)\rho_W(g)^{-1} = \Phi(T)$, and multiplying on the right by the invertible $\rho_W(g)$ gives the second claim. $\square$

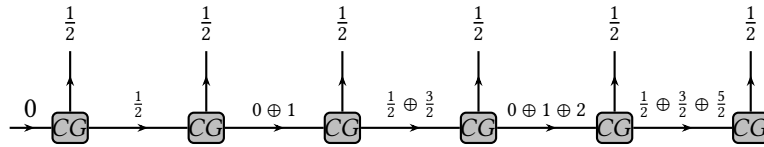
Hence $(V \otimes W^*)^G \cong \text{Hom}_G(W, V)$, and we identify T with $\Phi(T)$. For $W = V$ this is the familiar statement $U(g)HU(g)^{-1} = H \iff [H, U(g)] = 0$.



Corollary 3.3.3 (Networks of symmetric tensors are symmetric). *Since a tensor network is built by composition and application of the (co)ev maps, both of which are intertwiners, the full network is also an intertwiner. Important there is that this entire discussion is only about FdVect or FdHilb for now, but i think the same applies to any braided category.*

As an application of this more general tensor formalism, we can construct a symmetric MPS. If we stay with our current example, our ground state is of the form $|i_1 i_2 \cdots i_n\rangle \in (\mathbb{C}^2)^n$. If we represent this as a **MPS**, we know by default that this needs a physical dimension of 2, i.e. the space corresponding to the $\frac{1}{2}$ irrep at every site.

We can also use the presentations of the clebsch-gordan operators and realize that these can be considered tensors of order 3. If we chain these together naively we will get a state that can be represented symbolically as



and we can immediately observe an increasing bond dimension to the right.

Recall that a single mps tensor that is also symmetric is of the form $\text{Hom}_G(D_i, P_i \otimes D_{i+1})$. Note that we did not assume that P or D are irreps, therefore we can decompose such an MPS as

$$\text{Hom}_G \left(\bigoplus_i \mathbb{C}^{N_i^D} \otimes D_i, \left(\bigoplus_j \mathbb{C}^{N_j^P} \otimes P_j \right) \otimes \left(\bigoplus_k \mathbb{C}^{N_k^E} \otimes E_k \right) \right).$$

Rearranging this equation yields

$$\bigoplus_{ijk} \underbrace{\text{Hom}(\mathbb{C}^{N_i^D}, \mathbb{C}^{N_j^P} \otimes \mathbb{C}^{N_k^E})}_{3\text{legged tensor}} \otimes \underbrace{\text{Hom}_G(D_i, P_j \otimes E_k)}_{\text{Clebsch-Gordan tensor}}$$

Again, we know the physical space to be $\frac{1}{2}$, so we may simplify this expression to

TODO

We can do some analysis on the individual tensors here. If we choose the third tensor, of the form $\text{Hom}_G(0 \oplus 1, \frac{1}{2} \otimes (\frac{1}{2} \oplus \frac{3}{2}))$, we can do some analysis over the CG-rule for $\text{SU}(2)$.

There are 3 possible combination in the basis of this space, these are the spaces

$$(0, \frac{1}{2}, \frac{1}{2}); \quad (1, \frac{1}{2}, \frac{1}{2}); \quad (1, \frac{1}{2}, \frac{3}{2})$$

$$D^{(1; \frac{1}{2}, \frac{3}{2})} \in V^{\deg} 1 \otimes V^{\deg} \frac{1}{2} \otimes V^{\deg}_{\frac{3}{2}}.$$

Theorem 3.3.4 (Fundamental Theorem of MPS). *If two MPS states generated by A and B respectively. Figure out what we can do with this.*

Proof [[Cir+17](#)], [[VFW18](#)], [[Per+08](#)].

3.3.1 Algorithms for block-sparse tensors

ANYONS I : INTRODUCTION

This chapter introduces the theory of anyons by a set of physical requirements. The algebraic definition of anyons will then be given by the discussion in Chapter 5.

Anyons are proposed as a *defect* in a 2 dimensional space (manifold? compact?) with *exotic exchange statistics*. These exchange statistics differ from bosons or fermions in that we allow particles to pick up a arbitrary phase $e^{i\theta}$ on exchange in the abelian case, and a phase as a matrix element in the nonabelian case.

We group anyons into theories. An anyon theory carries a finite set of charges (types) $a, b, c \dots$ including a trivial *vacuum* charge. We are then interested in their spacetime histories, as this is where the topological properties of the anyon become apparent. First fix an oriented manifold D of dimension two as our base space. Then introduce an anyonic defect as a point like hole (that is also oriented), that is carrying a charge. Such a particles spacetime history is then given by an appropriate 3-manifold.

I dont like this whole section.

4.1 CONFIGURATION SPACES & THE BRAID GROUP

The original article by Leinaas and Myrheim [LM77] identifies a generalized theory of *identical* particles. Denote the space where a particle lives by Σ and thus the space of n particles possible coordinates of equal type by $\Sigma^n = \times^n \Sigma$. Since the particles are assumed to be identical, any permutation (an action of S_n) necessarily leads to the same physical state, so we denote the full space of our system by the quotient

$$\Sigma^n / S_n.$$

However we also want no two particles occupying the same space¹, so the diagonal in $\Delta = \{x \in \Sigma^n \mid x_i = x_j \text{ for some } i \neq j\}$ needs to be excluded as well [Wu84]. This leads to the expression

rephrase!

$$\text{Conf}_n(\Sigma) = (\Sigma^n - \Delta) / S_n$$

which is defined to be the *configuration space* of n unordered particles. However as we will see, anyons come in different "types", and we consider only particles of the same type to be identical. Later publications however define a swath of different notions of configuration spaces. We follow [SS22] with the following definition:

original publications.

Definition 4.1.1. The *ordered Configuration Space* of n particles of possibly different types (or labeled particles with the same type) on a surface of dimension D is

$$\text{Conf}_{\{1, \dots, n\}}(\Sigma) := (\Sigma)^n - \Delta_{\Sigma}^n$$

and the *unordered Configuration Space* of n unlabeled identical particles is defined as

$$\text{Conf}_n(\Sigma) := \text{Conf}_{\{1, \dots, n\}}(\Sigma) / S_n.$$

Note that while [LM77] conjectured computations for strange exchange statistics in both 2 and 3 dimensions, the concept of an anyon as a 1 dimensional representation is due to [Wil90].

¹If this were the case, then the resulting exchange statistics would be bosonic [Wu84, Footnote 12].

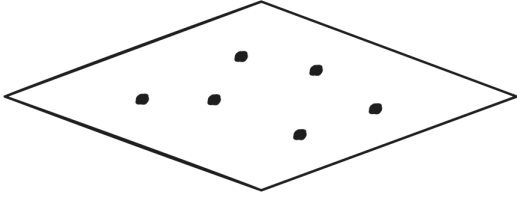


Figure 4.1: The configuration space of 5 particles in the plane $\mathbb{R}^2$.

Now we want to classify all the possible motions particles can take in this space. The *motion group* of the configuration space is given by the spaces *homotopy classes* (cf. Appendix A.2), which are given by the fundamental group(oid) of the space. Informally, this group collects all the loops at some configuration x_0 up to homotopy. We want to differentiate two cases here: If $D = 2$ it is known that

$$\pi_1(\text{Conf}_n(\Sigma)) = \text{Br}_n(\Sigma) \quad \text{and} \quad \pi_1(\text{Conf}_{\{1, \dots, n\}}(\Sigma)) = \text{PBr}_n(\Sigma)$$

however if $D \geq 3$ we know that both of these coincide with the symmetric group S_n instead. In fact there is an exact sequence

$$1 \longrightarrow \text{PBr}_n(\Sigma) \longrightarrow \text{Br}_n(\Sigma) \longrightarrow S_n \longrightarrow 1.$$

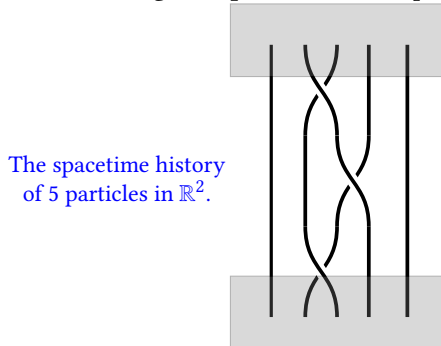
For an example in the plane $\mathbb{R}^2$ (or $\mathbb{C}^?$), we give an explicit representation of the *braid group* $\text{Br}_n(\mathbb{R}^2)$ and the *pure braid group* $\text{PBr}_n(\mathbb{R}^2)$. There is a well known presentation in terms of generators due to [artin25] as

$$\text{Br}_n(\mathbb{R}^2) = \left\langle \sigma_1, \dots, \sigma_{n-1} \left| \begin{array}{l} \sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \\ \sigma_i \sigma_j = \sigma_j \sigma_i, \quad |i - j| \geq 2 \end{array} \right. \right\rangle.$$

These generators also have a graphical interpretation as

$$\sigma_i := \left[\begin{array}{cccc} \cdots & | & \text{X} & | & \cdots \\ & \text{green } i-1 & \text{green } i & \text{green } i+1 & \text{green } i+2 \end{array} \right] \quad (4.1)$$

Now we can give a process as the *spacetime history* diagram of the n particles.



We can also do an example on the 2-sphere, since there is an extra relation of the $\mathbb{R}^2$ braid group. Generally, this braid group arises as a *quotient* of the Artin group, but this does not hold in general.

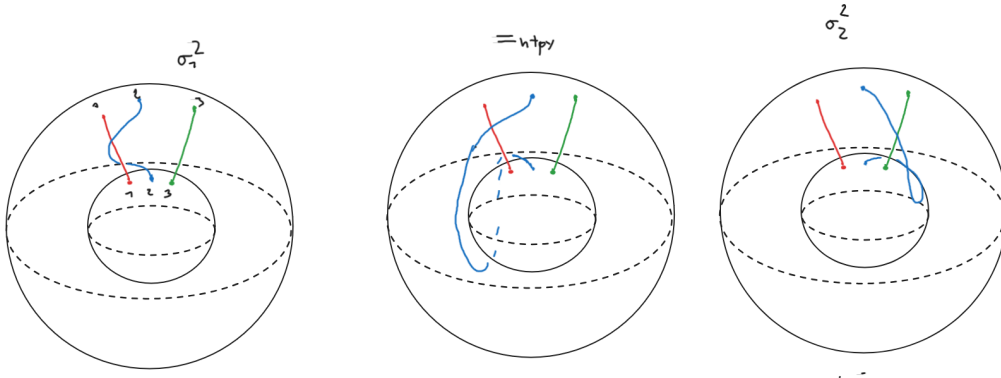


Figure 4.2: 3 Anyons on the sphere S^2 with the equivalence of the braids σ_1^2 and σ_2^2 . From an exercise [github](#) in [Sim23].

Analyzing this case further yields that

$$\pi_1(\text{Conf}_3(S^2)) \cong Br(S^2) \cong \mathbb{Z}_4 \boxtimes \mathbb{Z}_3$$

and thus finite. In fact, we know about the braid group on spheres that the order of $Br_n(S^2)$ is given by

$$|Br_n(S^2)| = \begin{cases} |\mathbb{Z}_2| = 2 & \text{if } n = 2, \text{ [FB62, p. 255]} \\ |???| = 12 & \text{if } n = 3, \text{ [nLa26, Prop 2.2]} \\ \infty & \text{if } n \geq 4. \end{cases} \quad (4.2)$$

Now, if we associate a Hilbert space to this configuration space, we can achieve this by using a *Hilbert Bundle*

$$\mathcal{H} \xrightarrow{\pi} \text{Conf}_{\{1, \dots, n\}}(\Sigma),$$

i.e. to each point $x \in \text{Conf}_{\{1, \dots, n\}}(\Sigma)$ there is an associated Hilbert space $\mathcal{H}_x$. Furthermore, a path through the configuration space $[\gamma] : x_0 \rightarrow x_1 \in \text{Conf}_{\{1, \dots, n\}}(\Sigma)$ is lifted to some unitary representation

$$\rho([\gamma]) \in U(\mathcal{H}_{x_0} \rightarrow \mathcal{H}_{x_1})$$

by *parallel transport*.

The closed paths in the respective groups form a *monodromy* in the configuration space and a *holonomy* in the Hilbert Bundle, and it is these generators that yield us unitary gates that are universal for *topological quantum computation* (cf. Section 6.3)

However, in the usual literature anyons are also identified over their *fusion tensor category* (cf. Chapter 5). Shifting the viewpoint from configuration spaces to fusion categories requires additional postulates about the systems we want to reason about. Although this correspondence is well known and the categorical interpretation can be considered standard, a formal treatment and proof were only given recently. These are pointed out in [Val21a], and formally rigorous derivations of this fact are given in [SS23]. However we will not give a derivation of this result and instead assume it. Perhaps a more natural introduction to this correspondence is given by an excursion over MCGs as done in [Bur16].

is this true?

Check if htis is a full proof.

CATEGORICAL NETWORKS

In the existing literature it is well known that all of the concepts we have seen up to now are modeled by a formalism called *tensor categories*:

1. fdVect_k and fdHilb tensors are the most trivial and motivating case and trivial build tensor categories.
2. $\text{Rep}(G)$ (for G finite) as in chapter 3 are one of the primary motivations for construction fusion tensor categories. [ENO17].
3. All anyonic topological orders (i.e species of quasiparticles as in section 4.1) can be classified by *modular tensor categories* (MTCs) [Kit06][RSW09].

Literature 5.0.1 (Tensor Categories). Grothendieck investigated what we now know as tensor categories as part of a larger study on *tannakian categories* to related the methods of representation theory (c.f. 3) to less related mathematical problems. Monoidal categories in turn were introduced by Benabou [empty citation] and MacLane [empty citation] under the name *bicategory* and *category with multiplication*, although an elementary construction of such categories is given by Tannaka [empty citation]

The name *tensor category* is due to Deligne [Del90] in "Catégories Tannakiennes" and its followup [Del02] BlaBlub

The correspondence between the penrose notation (i.e. planar diagrams) and the calculus provided by tensor categories is explained in great detail in Joyal and Street [JS91], although earlier accounts such as [RT91] already make use of such a string calculus.

We have also seen a method for evaluating networks built in $\text{Rep}(G)$ in chapter 3 by decomposing a network into its fusion tree and its degeneracy spaces. As an immediate consequence, this yields an algorithm for evaluating generalized *anyonic* and *categorical* tensor networks, as in [Unf24, § 4][DH25] [Pfe12; Pfe+10] [Aye17] [Sch+20, § ?].

In light of this we provide a bottom-up construction of the theory of fusion categories in section 5.1, each concept will carry along an example for representations, a physical interpretation in terms of anyonic world-lines as well as the correspondence to the ATS systems of Penrose [Pen71]. We formalize the aforementioned decomposition in section 5.2 and list the algorithms necessary for achieving a tensor network out of these theories.

If we however mention ATSs, it must be said that there is a multitude of tensor-like string calculi and correspondences to categories. A great review of these string like languages is given by Selinger [Sel11]. Furthermore, the string calculus as defined by penrose for the ATSs does not have an abstract formulation that is sound and coherent as defined by [JS91]. The definitions of these terms will be given in the boxes labeled Graphical Notation.

5.1 TENSOR & FUSION CATEGORIES

The general idea of a category is that it admits a sort of *interface*. As noticed, there are many similarities in data and structure between the entities we already dealt with, i.e. the tensor product of vector space, the tensor product of representations, of Hilbert spaces etc.... We can view all these concepts as a category with additional constraints added.

there is one here from 1982 find what this does.

or tqfts?

Definition 5.1.1. A *category* C consists of

- a set of *objects* $\text{ob}(C)$,
- for every pair $A, B \in \text{ob}(C)$, a *hom-set* $\text{Hom}_C(A, B)$ of *morphisms* $f : A \rightarrow B$,
- for every $A \in \text{ob}(C)$, an *identity morphism* $\text{id}_A : A \rightarrow A$,
- for every triple $A, B, C \in \text{ob}(C)$, a *composition map*

$$\circ : \text{Hom}_C(B, C) \times \text{Hom}_C(A, B) \longrightarrow \text{Hom}_C(A, C),$$

such that composition is *associative*, i.e. $(f \circ g) \circ h = f \circ (g \circ h)$ whenever defined, and *unital*, i.e. $f \circ \text{id}_A = f = \text{id}_B \circ f$ for all $f : A \rightarrow B$.

Example 5.1.2. Some examples here:

1. The category fdVect_k is the category of finite dimensional vector spaces and its morphisms are the linear operators between vector spaces.
2. The category fdHilb is the category whose objects are Hilbert spaces and whose morphisms are the bounded linear operators between them. Here we can alternatively take the objects to be the tuple $(\mathcal{H}, \langle -, - \rangle)$.
3. The category Set admits sets as objects and functions between sets as the morphisms.
4. The product category $C \times \mathcal{D}$ holds as objects tuples $(a \in C, b \in \mathcal{D})$ and as morphisms tuples of maps $(f \in \text{Hom}_C(a_1, a_2), g \in \text{Hom}_{\mathcal{D}}(b_1, b_2))$.

We can take categories and associate them with each other by use of a *functor*.

Definition 5.1.3. A *functor* F between the categories C and $\mathcal{D}$, denoted as $F : C \rightarrow \mathcal{D}$ assigns to each object $A \in C$ an object $F(A) \in \mathcal{D}$ as well as to every morphism in $f : A \rightarrow B \in \text{Hom}_C$ a morphism $F(f) : F(A) \rightarrow F(B) \in \text{Hom}_{\mathcal{D}}$. Additionally, this must satisfy

$$F(\text{id}_A) = \text{id}_{F(A)}$$

as well as one of the cases

$$F(g \circ f) = F(g) \circ F(f) \quad \text{in the } \textit{covariant} \text{ case} \quad (5.1)$$

or

$$F(g \circ f) = F(f) \circ F(g) \quad \text{in the } \textit{contravariant} \text{ case.} \quad (5.2)$$

We can also relate functor with each other through maps between them. An important relation here will be *naturality*:

Definition 5.1.4. A transformation τ between functors $F, G : C \rightarrow \mathcal{D}$ is *natural* if the diagram commutes for all $A : \text{ob}(C)$ and $f : \text{Hom}_C(A, B)$:

$$\begin{array}{ccc} F(A) & \xrightarrow{\tau_A} & G(A) \\ \downarrow F(f) & & \downarrow G(f) \\ F(B) & \xrightarrow{\tau_B} & G(B) \end{array}$$

Example 5.1.5. As an immediate application, consider the category of Hilbert spaces fdHilb and assign it a *forgetfull* functor $\omega : \text{fdHilb} \rightarrow \text{fdVect}_k$ that takes each Hilbert space to its underlying vector space, and each bounded linear operator to its linear operator between vector spaces.

We will now explore how to use category theory to model quantum processes. We want this to be able to do a list of things. First, given two systems A and B , we want the combined system to be an object as well. Given processes, i.e. operators on each individual system $f : A \rightarrow A', g : B \rightarrow B'$, there should be a combined process. Additionally, structures such as entanglement and no-cloning should be readily apparent in this theory.

Consider the tensor product $\otimes_{\mathbb{C}}$ of vector spaces V, W . We can define this using a *universal property*. This should motivate the following definition:

Definition 5.1.6. A *monoidal category* C is a category that admits a unit object $I \in ob(C)$ and a bifunctor $\otimes : C \times C \rightarrow C$, as well as 3 constraining isomorphisms

$$\rho_A : A \otimes I \cong A \quad \lambda_B : I \otimes B \cong B \quad \alpha_{ABC} : (A \otimes B) \otimes C \cong A \otimes (B \otimes C)$$

such that the following *triangle* and *pentagon*¹ diagrams commute:

$$\begin{array}{ccc} (A \otimes I) \otimes B & \xrightarrow{\alpha_{A,I,B}} & A \otimes (I \otimes B) \\ \rho_A \otimes id_B \searrow & & \swarrow id_A \otimes \lambda_B \\ & A \otimes B & \end{array} \quad \begin{array}{ccc} ((A \otimes B) \otimes C) \otimes D & \xrightarrow{\alpha_{A,B,C} \otimes id_D} & (A \otimes (B \otimes C)) \otimes D \\ \alpha_{A \otimes B, C, D} \downarrow & & \downarrow \alpha_{A, B \otimes C, D} \\ (A \otimes B) \otimes (C \otimes D) & & A \otimes ((B \otimes C) \otimes D) \\ \alpha_{A, B, C \otimes D} \searrow & & \swarrow id_A \otimes \alpha_{B, C, D} \\ & A \otimes (B \otimes (C \otimes D)) & \end{array}$$

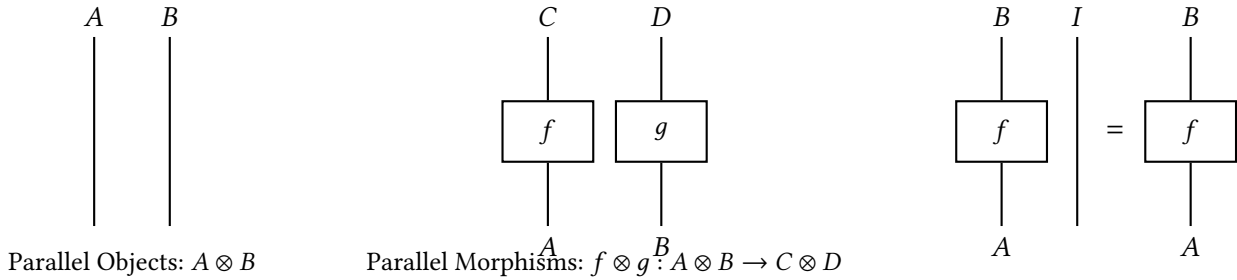
For simplicity we denote a monoidal category as the sextuple $(C, \otimes, I, \alpha, \lambda, \rho)$.

Note that because we defined $- \otimes - : C \times C \rightarrow C$ to be a *(bi)functor*, we immediately gain the following relations by functoriality:

$$(f \otimes g) \circ (f' \otimes g') = (f \circ f') \otimes (g \circ g') \quad (5.3)$$

$$id_{A \otimes B} = id_A \otimes id_B$$

Graphical Notation 5.1.7. We denote the elements of this Category by simple lines, and the morphisms as boxes between these lines. Since the tensor product corresponds to parallel execution, we describe tensored objects as parallel lines and tensored operators as parallel squares.



The commuting condition in definition 5.1.6 is an instance of the *MacLane-coherence-theorem* [maclane?] (for monoidal categories). This essentially asserts that no matter how many objects we look at, their tensoring are isomorphic to each other without specifying another isomorphism on higher spaces, but instead reusing the α associator on 3 strands is enough.

Example 5.1.8. Some examples of this tensor product here:

1. In fdVect_k , we denote the tensor product as $\otimes_k$, and this is the default kronecker-product on k -vector spaces. The monoidal unit in this case is the field k .

¹ Medieval mathematicians used to include the diagonal composite morphisms in this diagram, however, after a few stake burning incidents, this practice was abandoned.

Do we need to give an intro to these properties?

Does k need to be algebraically closed?

2. The category $\text{Rep}(G)$ is monoidal and has a tensor product by definition 3.1.5:

$$(V \otimes_k W, \rho_{V \otimes_k W})$$

and on morphisms $f \in \text{Hom}_G(V_1, V_2), g \in \text{Hom}_G(W_1, W_2)$ this gives

$$(f \otimes g)(v \otimes w) = f(v) \otimes g(w) \in \text{Hom}_G(V_1 \otimes_k W_1, V_2 \otimes_k W_2)$$

3. Take for example the Fib anyon theory. We can take products of anyons $a \otimes b$ and we can also identify the vacuum type I over $a \otimes I = a$ and so on. However we also know that these anyons come in pairs, so they ought to be in a fusion channel. For now we do not have a concept of antiparticles or of fusion channels. We can only give the pairings of the anyons by bracketings of the monoidal tensors.

rephrase!

Definition 5.1.9. Let $(C, \otimes_C, 1_C)$ and $(D, \otimes_D, 1_D)$ be monoidal categories. A *monoidal functor* is a functor $F : C \rightarrow D$ that preserves the monoidal structure

Definition 5.1.10. Let $(\mathcal{V}, \otimes_{\mathcal{V}}, I, \alpha, \lambda, \rho)$ be a monoidal category. A category C *enriched over $\mathcal{V}$* (or a $\mathcal{V}$ -category) consists of

1. for every $A, B \in \text{ob}(C)$, a *hom-object* $\text{Hom}_C(A, B) \in \mathcal{V}$ (as opposed to Set in normal category),
2. for every triple $A, B, C \in \text{ob}(C)$, a *composition morphism* in $\mathcal{V}$

$$\circ_{A,B,C} : \text{Hom}_C(B, C) \otimes_{\mathcal{V}} \text{Hom}_C(A, B) \longrightarrow \text{Hom}_C(A, C),$$

3. for every $A \in \text{ob}(C)$, a *unit morphism* in $\mathcal{V}$

$$j_A : I \longrightarrow \text{Hom}_C(A, A),$$

This immediately gives us another definition:

$$\text{Hom}_C(-, X) : C \rightarrow \mathcal{V} \quad \text{is a contravariant functor.}$$

Definition 5.1.11 ([JS91]). A *braided monoidal category* is a monoidal category $(C, \otimes, I, \alpha, \lambda, \rho)$ equipped with an additional isomorphism, the *braiding*

$$\beta_{A,B} : A \otimes B \cong B \otimes A,$$

on A and B satisfying

$$\beta_{A, B \otimes C} = (\text{id}_A \otimes \beta_{A,C}) \circ (\beta_{A,B} \otimes \text{id}_C) \quad \text{and} \quad \beta_{A \otimes B, C} = (\beta_{A,C} \otimes \text{id}_B) \circ (\text{id}_A \otimes \beta_{B,C}). \quad (5.4)$$

As a consequence, the *hexagon* relations commute for all $A, B, C \in C$:

$$\begin{array}{ccc} (A \otimes B) \otimes C \xrightarrow{\alpha} A \otimes (B \otimes C) \xrightarrow{\beta_{A, B \otimes C}} (B \otimes C) \otimes A & A \otimes (B \otimes C) \xrightarrow{\alpha^{-1}} (A \otimes B) \otimes C \xrightarrow{\beta} C \otimes (A \otimes B) \\ \beta_{A, B} \otimes \text{id}_C \downarrow & \downarrow \alpha & \downarrow \beta & \downarrow \alpha^{-1} \\ (B \otimes A) \otimes C \xrightarrow{\alpha} B \otimes (A \otimes C) \xrightarrow{\text{id}_B \otimes \beta_{A,C}} B \otimes (C \otimes A) & A \otimes (C \otimes B) \xrightarrow{\alpha^{-1}} (A \otimes C) \otimes B \xrightarrow{\beta} (C \otimes A) \otimes B \end{array}$$

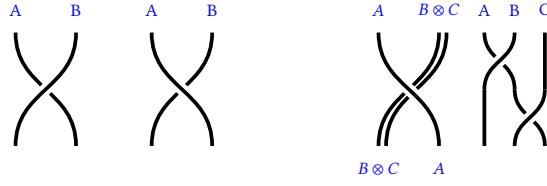
Definition 5.1.12. A braided monoidal category is *symmetric* if $\beta_{B,A} \circ \beta_{A,B} = \text{id}_{A \otimes B}$ for all A, B .

Lemma 5.1.13. The Yang-Baxter equation holds for the braiding:

$$(\beta_{A,B} \otimes \text{id}_t \text{ ext } C)(\text{id}_A \otimes \beta_{A,C})(\beta_{B,C} \otimes \text{id}_A) =$$

define this somewhere

Graphical Notation 5.1.14. Now the graphical calculus changes, as we need to consider whether the braidings defined by β are over or under. We denote the simple braidings as



Check for the correct definition of the ambient space here.

Example 5.1.15. 1. In fdVect_k this mapping is trivial.

2. For representations, there exist transformations on the basis states, for example the theory $\text{Rep}(\text{SU}(2))$ picks up the phase $(-1)^{j_a+j_b+j_c}$. For all categories built from representations of groups, these are symmetric.
3. For anyons, these correspond to the phase picked up when exchanging the anyons. For abelian anyons, these braidings are symmetric indeed, however for nonabelians these create non-symmetric categories.

Notation!

If we want to use contractions between matchings legs, and overall want the ability to take duals just like in FDCVec , we need both a concept of duality and the evaluation maps. This is also the categorification of the *creation* and *annihilation* maps from Chapter 6. Let this motivate the following definition:

Definition 5.1.16. A monoidal category $(C, \otimes, 1)$ is *rigid* if, for every $A \in C$, there exists a *dual object* A^* with the following maps:

$$\text{ev}_A : A^* \otimes A \rightarrow 1 \quad \text{and} \quad \text{coev}_A : 1 \rightarrow A \otimes A^*$$

subject to the diagrammatic constraints

$$\begin{array}{c} A \xrightarrow{\lambda_A} 1 \otimes A \xrightarrow{\text{coev}_A \otimes \text{id}_A} (A \otimes A^*) \otimes A \xrightarrow{\alpha_{A,A^*,A}} A \otimes (A^* \otimes A) \xrightarrow{\text{id}_A \otimes \text{ev}_A} A \otimes 1 \xrightarrow{\rho_A} A \\ \text{id}_A \end{array}$$

$$\begin{array}{c} A^* \xrightarrow{\rho_{A^*}} A^* \otimes 1 \xrightarrow{\text{id}_{A^*} \otimes \text{coev}_A} A^* \otimes (A \otimes A^*) \xrightarrow{\alpha_{A^*,A,A^*}} (A^* \otimes A) \otimes A^* \xrightarrow{\text{ev}_A \otimes \text{id}_{A^*}} 1 \otimes A^* \xrightarrow{\lambda_{A^*}} A^* \\ \text{id}_{A^*} \end{array}$$

Example 5.1.17. 1. In fdVect_k , the dual object for every vector space V is given by its *dual-space* $V^* := \text{Hom}(V, k)$. The evaluation and coevaluation maps here correspond to

$$\text{ev}_V : V \otimes V^* \rightarrow k,$$

$$\text{coev}_V : k \rightarrow V^* \otimes V,$$

2. In Anyon theories we generally want the ability to take antiparticles for each species of anyon, and furthermore, we want to be able to *create* and *annihilate* these anyons. So the dual object corresponds to the antiparticle, and the ev and coev maps either produce a particle-antiparticle pair from the vacuum or annihilate them (to a state containing the vacuum).

Graphical Notation 5.1.18. Now we have to introduce this notion of duality on the level of the graphical calculus. We denote an object in the category by an oriented arrow, and denote its dual object by the reverse.



Do these need to point up or down?

Similarly the ev_A and coev_A maps induce a graphical meaning:

$$\text{coev}_A : \begin{array}{c} A \\ \cup \\ A^* \end{array} \quad \text{and} \quad \text{ev}_A : \begin{array}{c} A^* \\ \cap \\ A \end{array}.$$

Observe that in the graphical calculus, we denoted this as a bent arrow from $A \rightarrow A^*$. In fact, we gain the following identities:

$$\text{Hom}_C(X \otimes A, Y) \cong \text{Hom}_C(X, A^* \otimes Y) : f \mapsto$$

We can also build dualities for morphisms here: Let $f : A \rightarrow B$ be arbitrary and define

$$f^* : B^* \rightarrow A^* = (\text{ev}_B \otimes \text{id}_{A^*})(\text{id}_{B^*} \otimes f \otimes \text{id}_{A^*})(\text{id}_{B^*} \otimes \text{coev}_A)$$

This gives us a few additional theorems:

1. $\text{id}_A^* = \text{id}_{A^*}$
2. $(g \circ f)^* = f^* \circ g^*$
3. $(f \otimes g)^* = g^* \otimes f^*$
4. $f \mapsto f^*$ is linear

Also notice by this that $(-)^*$ is a contravariant functor.

Further the natural observation that $A \cong A^{**}$ holds should be encoded in these categories.

Definition 5.1.19. A *pivotal structure* on a rigid monoidal category is given by a natural isomorphism $\text{id} \rightarrow (-)^{**}$, i.e. a collection of isomorphisms that identify $\{A \rightarrow A^{**}\}$.

This allows us to define a notion of a traced category.

Definition 5.1.20. We can define a notion of a *right trace* on any endomorphism $V \rightarrow V$ in such categories:

$$\text{tr}_R = 1 \xrightarrow{\text{coev}_A} A \otimes A^* \xrightarrow{(\delta_A \circ (-)) \otimes \text{id}_{A^*}} A^{**} \otimes A^* \xrightarrow{\text{ev}_{A^*}} 1$$

Definition 5.1.21. A pivotal rigid braided monoidal category is called *spherical* if

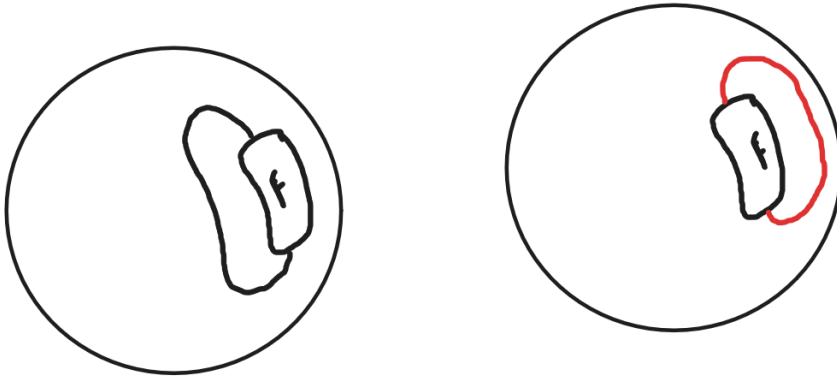
$$\text{tr}_R = \text{tr}_L$$

and we can further define the *quantum dimension* of an object A as

$$\dim(A) = \text{tr}(\text{id}_A) \tag{5.5}$$

Graphical Notation 5.1.22. The usage of *spherical* here derives from the fact that we can arbitrarily reorder our string calculus on the surface of the 2-sphere: Consider for example the trace where the connecting strand is swapped around the back of the sphere.

Does this lead naturally into the ribbon being the only obstacle to planarity?



Lemma 5.1.23. State without proof, in any spherical category C we can make the following statements about the trace. Let $A \in C$ and $f, g \in \text{End}(A)$ arbitrary, let

$$\text{tr} : \text{End}_C(A) \rightarrow \text{End}_C(I)$$

1. $\text{tr}(f^*) = \text{tr}(f)$
2. $\text{tr}(f \otimes g) = \text{tr}(f) \cdot \text{tr}(g)$
3. $\text{tr}(h \circ k) = \text{tr}(k \circ h)$
4. _____

Likely more, but which ones do we actually need?

So there is two reasons that we need this morphism here. One is to resolve the third Reidemeister move (which we havent introduced yet!) and another is that we can consider anyons to have an intrinsic rotation, however, this would contradict that we considered them equivalent particles with no degrees of freedom. I think [Bur16] uses a deformed braid group for this, but I would need to double check this.

Definition 5.1.24. A *ribbon* category is a braided monoidal rigid category $(C, \otimes, 1, \alpha, \gamma, \rho, \beta, \text{ev}, \text{coev})$ with maps $\theta_A : A \rightarrow A$ that realizes a *self-braiding* of the object. This is subject to

$$\theta_{A \otimes B} = (\theta_A \otimes \theta_B) \circ \beta_{B,A} \circ \beta_{A,B}$$

as well as

$$\theta_{A^*} = (\theta_A)^*$$

Example 5.1.25. Since this is an endomorphism, physically this acts by a phase.

Definition 5.1.26. Given a linear category C with objects $\{A_j\}_{j=1}^N \subseteq \text{ob}(C)$, an object $\bigoplus_{i=1}^n A_i \in C$ is called the *direct sum* if it is equipped with the injection and projection map

$$\iota_i : A_j \rightarrow \bigoplus A_i \quad \text{and} \quad \pi_i : \bigoplus A_j \rightarrow A_i$$

such that

$$\pi_i \circ \iota_i = \text{id}_{A_i} \quad \forall i \quad \text{and} \quad \sum_i \iota_i \pi_i = \text{id}_{\bigoplus A_i}$$

A great resource on semisimple categories [Bai17].

Definition 5.1.27. A *semisimple* category² is a category in which every object $S \in \text{ob}(C)$ decomposes as a finite direct sum

$$S \cong \bigoplus_i S_i, \quad S_i \text{ simple} \tag{5.6}$$

where a *simple* object is an object, where for any other object A that admits an injective map, the object is either the 0 morphism or an isomorphism to S . If we group these simples by isomorphism, we can restate (5.6) equivalently as

$$S \cong \bigoplus_i S_i^{\oplus n_i}$$

where $n_i = \dim \text{Hom}_C(S_i, S)$. Note that by our assumption of fdVect_k -enrichment, all the n_i s are finite. Further denote the set of simple objects by $\text{Irr}(C) \subseteq \text{ob}(C)$.

Example 5.1.28. 1. In $\text{Rep}(G)$, we know that either the representations are simple (Irreps) or they decompose as some direct sum representation.

2. In Anyonic theories, we primarily only consider simple objects $a, b, c \dots$ and tensor products between them. Then we assign each combination of anyons a fusion outcome $a \otimes b = c \oplus d \oplus \dots$.

Lemma 5.1.29 (Schur's lemma [Sch05]). If A and B are simple objects in a linear category over an algebraically closed field k (this is always $\mathbb{C}$ in our case), then it holds that

$$\text{Hom}(A, B) = \begin{cases} k \cdot \text{id}_S, & \text{if } A \cong B; \\ 0, & \text{if } A \not\cong B. \end{cases}$$

Proof [IR19, Theorem 7.5].

²There are multiple definitions of semisimplicity and semisimple categories, and here we are using the definition of *object simplicity*. A review of these definitions is given in [Bai17].

this is vector space intuition, not sure if this works. Maybe we need to introduce monomorphism? Need to review this definition for general categories.

Is this just the first part of schurs lemma?

Enrichment first.

Now assume that we take a semisimple category that is also monoidal. By semisimplicity, any of the objects of the form $R_a \otimes R_b$ are isomorphic to $\bigoplus_c R_c$ where $R_a, R_b, \{R_c\}$ are all simple. A simple consequence of Lemma 5.1.29 produces

$$\dim \text{Hom}_C(R_a \otimes R_b, R_c) = \dim \text{Hom}_C\left(\bigoplus_c N_d^{ab} R_d, R_c\right) = \dim \bigoplus_d \text{Hom}_C(R_d, R_c)^{\oplus N_d^{ab}} \stackrel{\text{lemma 5.1.29}}{=} \dim k^{N_c^{ab}}$$

Call this isomorphism $\Phi : \bigoplus_{c,\mu} R_c \rightarrow R_a \otimes R_b$. Then we can extend projection and injection maps in dependence of the tensor product by the following diagram:

$$\begin{array}{ccccccc} \bigoplus_c R_c & \xrightarrow{\iota_\mu} & \bigoplus_{\mu,c} R_c & \xrightarrow{\Phi} & R_a \otimes R_b & \xrightarrow{\Phi^{-1}} & \bigoplus_{\mu,c} R_c \xrightarrow{\pi_\mu} \bigoplus_c R_c \\ \uparrow \iota_c & \nearrow \iota_{c,\mu} & & \searrow \exists \iota_{c,\mu}^{ab} & & \searrow \exists \pi_{ab}^{c,\mu} & \downarrow \pi_c \\ R_c & & & & & & R_c \end{array} \quad (5.7)$$

Furthermore, these admit similar constraints as the direct sum:

$$\pi_{ab}^{c,\mu} \iota_{d,v}^{ab} = \pi_{c,\mu} \Phi^{-1} \Phi \iota_{d,v} = \delta_{cd} \delta_{\mu v} \text{id}_{R_c} \quad (5.8)$$

and

$$\sum_{c,\mu} \iota_{c,\mu}^{ab} \pi_{ab}^{c,\mu} = \Phi \left(\sum_{c,\mu} \iota_{c,\mu} \pi_{c,\mu} \right) \Phi^{-1} = \Phi \text{id}_{\bigoplus_c R_c} \Phi^{-1} = \text{id}_{R_a \otimes R_b} \quad (5.9)$$

We take these maps as above and claim they form a basis of the spaces $\text{Hom}_C(a \otimes b, c)$ and $\text{Hom}_C(c, a \otimes b)$.

Proof. First we show linear independence. Let $\{x_\mu\}_1^{N_c^{ab}}$ be a set of scalars and suppose that $\sum_\mu x_\mu \iota_{c,\mu}^{ab} = 0$. Observe that

$$\pi_{c,\mu}^{ab} \left(\sum_\mu x_\mu \iota_{c,\mu}^{ab} \right) = \sum_\mu x_\mu \delta_{\mu,v} \text{id}_{R_c} = x_v \text{id}_{R_c}$$

and since $\text{id}_{R_c} \neq 0$ these form a linearly independent set.

Now let $f \in \text{Hom}_C(R_c, R_a \otimes R_b)$ be an arbitrary morphism. By (5.9), we can write

$$f = \text{id}_{R_a \otimes R_b} \circ f = \sum_{d,v} \iota_{d,v}^{ab} (\pi_{ab}^{d,v} \circ f \in \text{Hom}_C(R_c, R_d)) \stackrel{\text{Lemma 5.1.29}}{=} \sum_{d,v} \iota_{d,v}^{ab} \delta_{R_c \cong R_d} \lambda_v \text{id}_{R_c} = \sum_v \lambda_v \iota_{c,v}^{ab}$$

and f is an arbitrary linear combination and thus

$$\text{span}_{\mathbb{C}} \left\{ \iota_{d,v}^{ab} \right\}_{v=1}^{N_c^{ab}} \cong \text{Hom}_C(R_c, R_a \otimes R_b)$$

and the statement follows dually for $\text{Hom}_C(R_a \otimes R_b, R_c)$. $\square$

If we look at Hom spaces of a higher order such as $\text{Hom}_C(R_d, (R_a \otimes R_b) \otimes R_c)$, we have to choose a bracketing of the tensor products since we can only associate $(R_a \otimes R_b) \otimes R_c \cong R_a \otimes (R_b \otimes R_c)$. Under the (covariant) functor $\text{Hom}_C(-, R_d)$ this introduces a pushforward

$$(\alpha_{abc}^d)^* : \text{Hom}_C((R_a \otimes R_b) \otimes R_c, R_d) \cong \text{Hom}_C(R_a \otimes (R_b \otimes R_c), R_d)$$

and a pullback under $\text{Hom}_C(R_d, -)$

$$(\alpha_d^{abc})^* : \text{Hom}_C(R_d, (R_a \otimes R_b \otimes R_c)) \cong \text{Hom}_C(R_d, R_a \otimes (R_b \otimes R_c)).$$

We can also define a basis for these extended spaces. In fact, lets generalize this to the space $\text{Hom}_C(R_b, \bigotimes_{i=1}^n R_{a_i})$ where we assume this is fully left-aligned bracketing.

We define the higher bases recursively: Let $n > 2$ and intermediate labels $e_1, e_2, \dots, e_{n-2} \in \text{Irr}(C)$ and multiplicity indices $\mu_1, \dots, \mu_{n-1}$. We fix some notation to let $\vec{\mu}_{k-1}$ to denote the tuple of all labels up to μ_{k-1} . Then we can define

$$\iota_{d, \vec{e}_{n-2}, \vec{\mu}_{n-1}}^{a_1 \dots a_n} := \left(\iota_{e_{n-2}, \vec{e}_{n-3}, \vec{\mu}_{n-2}}^{a_1 \dots a_{n-1}} \otimes \text{id}_{R_{a_n}} \right) \circ \iota_{d, \mu_{n-1}}^{e_{n-2} a_n}.$$

We can assume this holds by induction for $(n - 1)$:

$$\pi_{e;\vec{e},\vec{\mu}}^{a_1\cdots a_{n-1}} \circ \iota_{e';\vec{e}',\vec{\mu}'}^{a_1\cdots a_{n-1}} = \delta_{ee'} \delta_{\vec{e}\vec{e}'} \delta_{\vec{\mu}\vec{\mu}'} \text{id}_{R_e}.$$

$$\sum_{e, \vec{e}, \vec{\mu}} \iota_{e; \vec{e}, \vec{\mu}}^{a_1 \cdots a_{n-1}} \circ \pi_{e; \vec{e}, \vec{\mu}}^{a_1 \cdots a_{n-1}} = \text{id}_{\bigotimes_{i < n} R_{a_i}}.$$

$$\pi_{c;\vec{e},\vec{\mu}}^{a_1\cdots a_n} \circ \iota_{d,\vec{f},\vec{v}}^{a_1\cdots a_n} \stackrel{(5.3)}{=} \pi_{c,\mu_{n-1}}^{e_{n-2}a_n} \circ \left[\left(\pi_{e_{n-2};\vec{e}_{n-3},\vec{\mu}_{n-1}}^{a_1\cdots a_{n-1}} \circ \iota_{f_{n-2};\vec{f}_{n-3},\vec{v}_{n-1}}^{a_1\cdots a_{n-1}} \right) \otimes \text{id}_{R_{a_n}} \right] \circ \iota_{d,v_{n-1}}^{f_{n-2}a_n}.$$

By the inductive hypothesis the bracket evaluates to $\delta_{e_{n-2}f_{n-2}} \delta_{\vec{e}_{n-3}\vec{f}_{n-3}} \delta_{\vec{\mu}_{n-2}\vec{\nu}_{n-2}} \text{id}_{R_{e_{n-2}}}$, and the $n = 2$ case finishes orthoonality.

Let $e := e_{n-2}$, $l' = l_{e; \vec{e}_{n-3}, \vec{\mu}_{n-2}}^{a_1 \cdots a_{n-1}}$, and π' likewise. Then

$$\sum_{\substack{d, \vec{e}, \vec{\mu} \\ a_1 \cdots a_n}} l_{d, \vec{e}, \vec{\mu}}^{a_1 \cdots a_n} \circ \pi_{d, \vec{e}, \vec{\mu}}^{a_1 \cdots a_n} = \sum_{\substack{e, \vec{e}_{n-3}, \vec{\mu}_{n-2}}} (l' \otimes \text{id}_{R_{a_n}}) \circ \left[\sum_{d, \mu_{n-1}} l_{d, \mu_{n-1}}^{e a_n} \circ \pi_{d, \mu_{n-1}}^{e a_n} \right] \circ (\pi' \otimes \text{id}_{R_{a_n}}).$$

In fact, we can also form a "canoncial" basis for the spaces $\text{Hom}_C(X^{\otimes n}, X^{\otimes n})$ as given in [Val21b]

this needs a lot of conventions?

$$\text{Hom} \left(\bigotimes_{k=1}^m X_{i_k}, \bigotimes_{l=1}^n X_{j_l} \right) = \text{span}_{\mathbb{C}} \left\{ \begin{array}{c} \begin{array}{ccccccc} X_{i_1} & X_{i_2} & X_{i_3} & \dots & X_{i_{m-1}} & X_{i_m} \\ & \searrow & \downarrow & & \nearrow & \nearrow \\ & e_1 & & & & \\ & & \searrow & \ddots & & \\ & & & e_{m-3} & & \\ & & & & \nearrow & \nearrow \\ & & & & e_{m-2} & \\ & & & & & b \\ & & & & \nwarrow & \swarrow \\ & & & & f_{n-2} & \\ & & & & \nearrow & \\ & & & & f_{n-3} & \\ & & & & \vdots & \\ & & & & f_1 & \\ & & & & \nwarrow & \swarrow \\ & & & & f_2 & \\ & & & & \nearrow & \\ X_{j_1} & X_{j_2} & X_{j_3} & \dots & X_{j_{n-1}} & X_{j_n} \end{array} \\ b, e_1, \dots, e_{m-2}, f_1, \dots, f_{n-2} \\ \mu_1, \dots, \mu_{m-2}, v_1, \dots, v_{n-2} \end{array} \right\} \quad (5.10)$$

Now, what this construction allows us to do is as follows: Consider the decompositions

$$\mathrm{Hom}_C((A \otimes B) \otimes C, D) \cong \bigoplus_{e \in \mathrm{Irr}(C)} \mathrm{Hom}_C(A \otimes B, e) \otimes_k \mathrm{Hom}_C(e \otimes C, D)$$

$$\mathrm{Hom}_C(A \otimes (B \otimes C)) \cong \bigoplus_{f \in \mathrm{Irr}(C)} \mathrm{Hom}_C(A \otimes f, D) \otimes_k \mathrm{Hom}_C(B \otimes C, f)$$

Which is induced by inverse of the isomorphism

refer to basis

$$\mu \otimes_k v \mapsto (\mu \otimes \mathrm{id}_c) \circ v.$$

Using this we can coordinatize the pushforward as a matrix

$$F_D^{ABC} : \bigoplus_{e \in \text{Irr}(C)} \text{Hom}_C(A \otimes B, e) \otimes_k \text{Hom}_C(e \otimes C, D) \rightarrow \bigoplus_{f \in \text{Irr}(C)} \text{Hom}_C(A \otimes f, D) \otimes_k \text{Hom}_C(B \otimes C, f)$$

where the entries of this matrix are called the $6j$ -symbols after Wigner. However to actually get an element we need to specify 10 parameters³, they are determined on the basis vectors as such:

$$\iota_{e,\mu}^{ab} \otimes_k \iota_{f,\nu}^{ec} = \sum_{f,\alpha,\beta} [F_d^{abc}]_{(e\mu\nu)}^{(f\alpha\beta)} \iota_{d,\beta}^{af} \otimes_k \iota_{f,\alpha}^{bc} \quad (5.11)$$

and one should realize that this is really just a change of base matrix.

Graphically, we can represent this as

$$F_d^{abc} : \mathbb{C}^{\mu_0+\nu_0} \left[\begin{array}{c} a \quad b \quad c \\ \swarrow \quad \searrow \quad \swarrow \\ e_0 \quad \quad d \end{array} \right] \oplus \mathbb{C}^{\mu_1+\nu_1} \left[\begin{array}{c} a \quad b \quad c \\ \swarrow \quad \searrow \quad \swarrow \\ e_1 \quad \quad d \end{array} \right] \oplus \dots \xrightarrow{\cong} \mathbb{C} \left[\begin{array}{c} a \quad b \quad c \\ \swarrow \quad \searrow \quad \swarrow \\ \quad \quad f_0 \end{array} \right] \oplus \mathbb{C} \left[\begin{array}{c} a \quad b \quad c \\ \swarrow \quad \searrow \quad \swarrow \\ \quad \quad f_1 \end{array} \right] \oplus \dots$$

We can do the same construction for the braiding:

$$R_c^{ab} : \text{Hom}(R_a \otimes R_b, R_c) \rightarrow \text{Hom}(R_b \otimes R_a, R_c)$$

$$\iota_{c,\mu}^{ba} = \sum_{\nu} [R_c^{ab}]_{\mu}^{\nu} \iota_{c,\nu}^{ab}$$

Example 5.1.30. Lets look at some examples of semisimple monoidal categories.

1. We know that representations admit semisimple theories and are endowed with a tensor product. These can be considered enriches as the morphisms in $\text{Rep}()$ are just linear maps. In the example of $\text{SU}(2)$, the basis maps $\iota_{c,\nu}^{ab}$ are equal to the *Clebsch-Gordan* coefficients, in fact these are multiplicity free and thus we can omit the index ν .
2. In anyon theories however, these bases are fully symbolic. They denote the fusion and splitting spaces of an anyon.

CG-coeffs can be indexed here (they are not symbolic only!)

Now we would like the category that has all of these statements combined:

Definition 5.1.31. A *fusion-category* is a semisimple, fdVect_k -enriched rigid braided monoidal category, where each simple object has only *finitely many* isomorphism classes. Furthermore, we required that I (the monoidal unit) is simple, equivalently $\text{End}_C(I) \cong k$.

Remark 5.1.32. While each representation category $\text{Rep}(G)$ forms a fusion category for finite G , this is not the case for infinite G such as $\text{SU}(2)$ or $\text{U}(1)$. These violate the condition that there are finitely many simples. We can remedy this by instead considering a deformation of the underlying Lie-algebra as $U_q(\mathfrak{sl}_2)$. This has various reasons, one of them is that we only wish to consider gapped theories for our topological orders

From the discussion above we can synthesize an alternative but equivalent definition of a fusion category. This is the definition that is provided in the physics literatur such as in [Kit06] or [Bon07; BSS08].

Definition 5.1.33 ([Mos25]). A *fusion category* can be restated in its vector space form as a finite set of labels (sectors?) L and vector spaces $V_c^{ab} := \text{Hom}_C(a \otimes b, c)$ for all $a, b, c \in L$, as well as the isomorphisms

$$F_d^{abc} : \bigoplus_e V_e^{ab} \otimes V_d^{ec} \rightarrow \bigoplus_e V_d^{ae} \otimes V_e^{bc}$$

such that the pentagon holds:

$$\begin{array}{ccc} & \bigoplus_{e,l} V_e^{ab} \otimes V_f^{cd} \otimes V_g^{el} & \\ \text{\scriptsize } id_{V_e^{ab}} \otimes F_g^{ecd} \nearrow & & \searrow \text{\scriptsize } F_g^{abl} \otimes id_{V_l^{cd}} \\ \bigoplus_{e,f} V_e^{ab} \otimes V_f^{ec} \otimes V_g^{fd} & & \bigoplus_{k,l} V_g^{ak} \otimes V_k^{bl} \otimes V_l^{cd} \\ \text{\scriptsize } F_f^{abc} \otimes id_{V_g^{fd}} \searrow & & \nearrow \text{\scriptsize } F_k^{bcd} \otimes id_{V_g^{ak}} \\ \bigoplus_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} & \xrightarrow{\text{\scriptsize } F_g^{afd} \otimes id_{V_{bc}^{k,h}}} & \bigoplus_{k,h} V_g^{ak} \otimes V_h^{bc} \otimes V_k^{hd} \end{array} \quad (5.12)$$

³Wigner (on $\text{SU}(2)$) considered theories that were *multiplicity-free*, so 6 symbols were sufficient.

The unit objects are given by

So far we have arrived at the structure of a fusion category. While we do not have quantum systems based upon the objects of these categories, we can model some processes through the hom-spaces, and we have $\mathbb{C}$ -enrichment for them as well. To model quantum systems, we need them to be Hilbert spaces as well. Classically this is done by defining a *unitary structure* on the fusion category.

Definition 5.1.34. Let C be a fusion category over $\mathbb{C}$. A *unitary structure* on C is a family of maps

$$(-)^\dagger : \text{Hom}_C(A, B) \rightarrow \text{Hom}_C(B, A)$$

which is the identity on objects and satisfies

1. (*involution*) $(\lambda f + \mu g)^\dagger = \bar{\lambda} f^\dagger + \bar{\mu} g^\dagger$ and $f^{\dagger\dagger} = f$.
2. (*Contravariance*) $(f \circ g)^\dagger = g^\dagger \circ f^\dagger$ and $\text{id}_A^\dagger = \text{id}_A$.
3. (*Positivity*) $f^\dagger \circ f = 0 \implies f = 0$.
4. (*Monoidality*) $(f \otimes g)^\dagger = f^\dagger \otimes g^\dagger$, and the associator and unitors are unitary, e.g. $\alpha_{A,B,C}^\dagger = \alpha_{A,B,C}^{-1}$.
5. (*Duality*) the pivotal isomorphism $j_A : A \rightarrow A^{**}$ is unitary, equivalently $(f^\dagger)^* = (f^*)^\dagger$ for all f .

A fusion category together with this structure is called a *unitary fusion category*.

Proposition 5.1.35. In a unitary fusion category:

1. $\text{Hom}_C(A, B)$ is a finite-dimensional Hilbert space for all A, B .
2. The bases can be chosen with $\pi_{ab}^{c,\mu} = (\iota_{c,\mu}^{ab})^\dagger$, so each ι is an isometry, Φ is unitary, and $u^\dagger$ is an orthogonal projection. The higher fusion-tree bases are orthonormal.
3. The F -symbols F_d^{abc} are unitary matrices.
4. We can define an inner product on C by

$$\langle f, f \rangle = \text{tr}(f^\dagger \circ f)$$

5. Reciprocity $\text{Hom}(R_c, R_a \otimes R_b) \cong \text{Hom}(R_a^* \otimes R_c, R_b)$ is unitary up to the factor ???.
6. If C is braided with braiding β , then $\beta_{A,B}^\dagger = \beta_{A,B}^{-1}$, so the R -symbols are unitary. In the ribbon case the twist satisfies $\theta_A^\dagger = \theta_A^{-1}$.

Now there is only one last structure missing before we can model anyons, and this is perhaps the least intuitive of all of these structures. While foundational for the physical interpretations of these theories, it is of little interest to our tensor formalisms and we will not go into depth on this.

Definition 5.1.36. In a braided fusion category C with $A, B \in C$ define the *S-matrix*

$$S_{AB} := \text{tr}(\beta_{B,A} \circ \beta_{A,B}).$$

If S is invertible, we call the category *modular*.

So why can we not jst do a Hilber-Enrichment?

We could give this as a functor, but this has a weird target.

5.2 DECOMPOSITIONS FOR CATEGORICAL NETWORKS

This chapter generalizes the decomposition in [SV12, Equation 27] to both Fusion Categories and the Representation categories of compact groups. Note that in [DH25], such a generalization is given for representations of finite groups, which does not include $SU(2)$. However I do not yet know if we should do so, since I not yet sure how to argue the F moves here. If we instead take the deformation $U_q(sl_2)$ we get finiteness (and fusion cats?), unfortunately $U_q(sl_2)$ is not a group

Definition 5.2.1. Recall that in a semisimple category C we can express every object S as a finite direct sum over isomorphism classes

$$S \cong \bigoplus_i S_i^{\oplus n_i}$$

and we also work purely over $\mathbb{C}$ here. We now wish to mirror *isotypic decomposition* definition 3.1.7 within categories, i.e. we wish to categorify this construction. Let V be a vector space of dimension n , i.e. choose a basis $v_1, \dots, v_n$ and define

$$V \otimes C := \bigoplus_{i=1}^n C$$

and we call V the *multiplicity space* of the object $C \in C$.

Furthremore, we can assert that $- \otimes - : \text{fdVect}_k \times C \rightarrow C$ is a functor.

Lemma 5.2.2. Let C be a k -linear category and let D, E be finite-dimensional k -vector spaces. For any objects $R, S \in C$, there is a natural isomorphism

$$\text{Hom}_C(D \otimes R, E \otimes S) \cong \text{Hom}_k(D, E) \otimes \text{Hom}_C(R, S).$$

Proof. Let $m = \dim_k D$ and $n = \dim_k E$. Proof by isomorphism chains:

$$\begin{aligned} \text{Hom}_C(D \otimes R, E \otimes S) &\cong \text{Hom}_C(R^{\oplus m}, S^{\oplus n}) && \text{by ??} \\ &\cong \bigoplus_{i=1}^m \bigoplus_{j=1}^n \text{Hom}_C(R, S) && \text{by additivity of } \text{Hom}_C \\ &\cong k^{mn} \otimes \text{Hom}_C(R, S) && \text{by ??} \\ &\cong \text{Hom}_k(D, E) \otimes \text{Hom}_C(R, S) && \text{since } \text{Hom}_k(D, E) \cong k^{mn} \end{aligned}$$

(The last as s)

□

Theorem 5.2.3 (Decomposition of Semisimple Categories). Let C be a semisimple, monoidal, k -linear category whose tensor product is k -linear in each variable, with simple objects $\{R_a\}_{a \in L}$ and let V_i, W_i be a collection of (not necessarily simple) objects in C . Every morphism

$$T \in \text{Hom}_C \left(\bigotimes_{i=1}^n V_i, \bigotimes_{i=1}^m W_i \right)$$

admit a decomposition as

$$T \cong \bigoplus_{\hat{a}, \hat{b}} \text{Hom}_k(D_{\hat{a}}, E_{\hat{b}}) \otimes_k \text{Hom}_C(R_{\hat{a}}, R_{\hat{b}}),$$

where $\hat{a} = (a_1, \dots, a_n)$ and P is a k -vector space of dimension ???

Redo this over a universal property, this is not natural.

Proof.

$$\begin{aligned}
& \text{Hom}_C \left(\bigotimes_i V_i, \bigotimes_j W_j \right) \\
& \cong \text{Hom}_C \left(\bigotimes_i \left(\bigoplus_{a_i} D_{i,a_i} \otimes R_{a_i} \right), \bigotimes_j \left(\bigoplus_{b_j} E_{j,b_j} \otimes R_{b_j} \right) \right) \quad (??) \\
& \cong \text{Hom}_C \left(\bigoplus_{\hat{a}} D_{\hat{a}} \otimes R_{\hat{a}}, \bigoplus_{\hat{b}} E_{\hat{b}} \otimes R_{\hat{b}} \right) \quad (\text{linearity of } \otimes) \\
& \cong \bigoplus_{\hat{a}, \hat{b}} \text{Hom}_C(D_{\hat{a}} \otimes R_{\hat{a}}, E_{\hat{b}} \otimes R_{\hat{b}}) \quad (\text{additivity of } \text{Hom}_C) \\
& \cong \bigoplus_{\hat{a}, \hat{b}} \text{Hom}(D_{\hat{a}}, E_{\hat{b}}) \otimes \text{Hom}_C(R_{\hat{a}}, R_{\hat{b}}) \quad (\text{Lemma 5.2.2})
\end{aligned}$$

In a basis expansion this then looks like

$$\cong \bigoplus_{\hat{a}, \hat{b}} \sum_{e, f, \mu, \nu} \text{Hom}(D_{\hat{a}}, E_{\hat{b}}) \otimes_k \widehat{T}_{e, f; \mu, \nu}^{\hat{a}, \hat{b}}$$

□

Corollary 5.2.4 (Operators are block matrices [KZ22]). *thm header. An operator*

$$\text{Hom}(V, W) \cong \text{Hom}_C \left(\bigoplus_a R_a^{\oplus d_a}, \bigoplus_b R_b^{\oplus e_b} \right) \cong \bigoplus_{a, b} \text{Hom}(D_a, E_b) \otimes \text{Hom}_C(R_a, R_b)$$

and by Schurs lemma this is just

$$\bigoplus_{a \in \text{Irr}(V) \cap \text{Irr}(W)} \text{Hom}(D_a, E_a) \cong \bigoplus_{a \in \text{Irr}(V) \cap \text{Irr}(W)} \text{Mat}_{d_a \times e_a}(k)$$

Corollary 5.2.5. Let $Q_{\mu, \eta}$ be a basis of $\text{Hom}_C(\bigotimes_{\hat{a}} R_{\hat{a}}, \bigotimes_{\hat{b}} R_{\hat{b}})$ where μ denotes the multiplicities and η the internal fusions. In a multiplicity-free Category we will omit μ as a label.

Example 5.2.6. However, now assign $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ to each leg in the example above. Each leg then becomes $V_0 \oplus V_1 \cong \mathbb{C} \otimes V_0 \oplus \mathbb{C} \otimes V_1$. Then we can start with the decomposition.

$$\begin{aligned}
& \text{Hom}_C \left(\bigoplus_{a, b, c \in (0,1)} (C \otimes C \otimes C) \otimes V_a \otimes V_b \otimes V_c, \bigoplus_{b \in (0,1)} C \otimes V_b \right) \\
& = \bigoplus_{a, b, c, d \in (0,1)} \text{Hom}_C((C \otimes C \otimes C) \otimes (V_a \otimes V_b \otimes V_c), C \otimes V_d) \\
& = \bigoplus_{a, b, c, d} \text{Hom}(C^3, C) \otimes \text{Hom}_{\text{Rep}(SU(2))}(V_a \otimes V_b \otimes V_c, V_d)
\end{aligned}$$

Example 5.2.7. Easier. However, now assign $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ to each leg in the example above. Each leg then becomes $V_0 \oplus V_1 \cong \mathbb{C} \otimes V_0 \oplus \mathbb{C} \otimes V_1$. Then we can start with the decomposition.

$$\begin{aligned}
& \text{Hom}_C \left(\bigoplus_{a, b \in (0,1)} (C \otimes C) \otimes V_a \otimes V_b, \bigoplus_{b \in (0,1)} C \otimes V_b \right) \\
& = \bigoplus_{a, b, c \in (0,1)} \text{Hom}_C((C \otimes C) \otimes (V_a \otimes V_b), C \otimes V_c) \\
& = \bigoplus_{a, b, c} \text{Hom}(C^2, C) \otimes \text{Hom}_{\text{Rep}(SU(2))}(V_a \otimes V_b, V_c)
\end{aligned}$$

Example 5.2.8 (XXZ-Hamiltonian). Consider a 4 leg tensor in $\text{End}(V_{1/2} \otimes V_{1/2})$, this decomposes as

$$A$$

Example 5.2.9. Lets look at a tensor with in legs $(\tau \otimes \tau)$ and τ and out legs τ .

$$\text{Hom}_C(((k \otimes 1) \oplus (k \otimes \tau)) \otimes (k \otimes \tau), k \otimes \tau) = \text{Hom}_C()$$

$$\begin{aligned} & \text{Hom}_C(((k \otimes 1) \oplus (k \otimes \tau)) \otimes (k \otimes \tau), k \otimes \tau) \\ & \cong \text{Hom}_C((k \otimes 1 \otimes k \otimes \tau) \oplus (k \otimes \tau \otimes k \otimes \tau), k \otimes \tau) \\ & \cong \text{Hom}_C(k \otimes 1 \otimes k \otimes \tau, k \otimes \tau) \oplus \text{Hom}_C(k \otimes \tau \otimes (k \otimes \tau), k \otimes \tau) \\ & \cong \text{Hom}(k \otimes k, k) \otimes \text{Hom}_C(1 \otimes \tau, \tau) \oplus \text{Hom}(k \otimes k, k) \otimes \text{Hom}_C(\tau \otimes \tau, \tau). \end{aligned}$$

Example 5.2.10 (Anyonic matrix operator). Finally, consider a $4 \rightarrow 4$ anyon operator with one leg on each side, representing the space $\text{Hom}((\tau \otimes \tau \otimes \tau \otimes \tau), (\tau \otimes \tau \otimes \tau \otimes \tau))$. The full decomposition is $2 + 3\tau$. Hence we obtain

$$\begin{aligned} & \text{Hom}_C((k^2 \otimes 1) \oplus (k^3 \otimes \tau), (k^2 \otimes 1) \oplus (k^3 \otimes \tau)) \\ & = \bigoplus_{a,b \in (1,\tau)} \text{Hom}_C(k^{m_a} \otimes a, k^{m_b} \otimes b) = \bigoplus_{a,b \in (1,\tau)} \text{Hom}(k^{m_a}, k^{m_b}) \otimes \text{Hom}_C(a, b) \end{aligned}$$

where

$$m_1 = 2, \quad m_\tau = 3.$$

For this case, Schurs lemma applies. So we have

$$= \text{Hom}(k^2, k^2) \oplus \text{Hom}(k^3, k^3)$$

which is just a block matrix with indexing $(1_1, 1_2, \tau_1, \tau_2, \tau_3)$.

5.2.1 The Deligne Tensor Product & Product Theories

$$\begin{array}{ccc} & \bigoplus_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} & \\ \nearrow & & \searrow \\ \bigoplus_{e,f} V_e^{ab} \otimes V_f^{ec} \otimes V_g^{fd} & & \bigoplus_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} \\ \downarrow F_f^{abc} \otimes \text{id}_{V_g^{fd}} & & \uparrow \\ \bigoplus_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} & \xrightarrow{F_f^{abc} \otimes \text{id}_{V_g^{fd}}} & \bigoplus_{k,h} V_g^{ak} \otimes V_h^{bc} \otimes V_k^{hd} \end{array}$$

ANYONS II: SPIN CHAINS & TENSOR NETWORKS

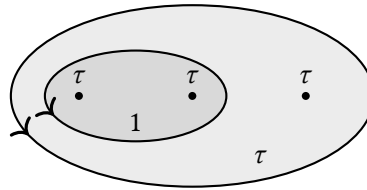
This chapter the diagrammatic methods that yield tensor networks on physical systems that constitute *anyons*. We introduce the fundamental theory of anyons algebraically (Section 6.1), and make use of the algebraic rules as the building blocks for our theory. Section 6.1.1 gives examples of how the statistical properties of anyons change with respect to it's ground state. Then, in Section 6.2, we provide a worked examples of how anyons are treated on *spin-chains*, which motivates some aspect of the tensor networks approach. TODO tensor network chapter. One of the primary motivations for the study of anyons are their universality for quantum computation. In ?? we provide an interpretation of how both normal braiding circuits and a variation in terms of *weaves* can be simulated by the tensor network theory.

6.1 ALGEBRAIC THEORY OF ANYONS

Anyons are proposed as a *defect* in a 2 dimensional space (manifold? compact?) with exotic exchange statistics. These exchange statistics differ from fermions in that we allow particles to pick up a arbitrary phase $e^{i\theta}$ on exchange in the abelian case, and a phase in a matrix element in the nonabelian case.

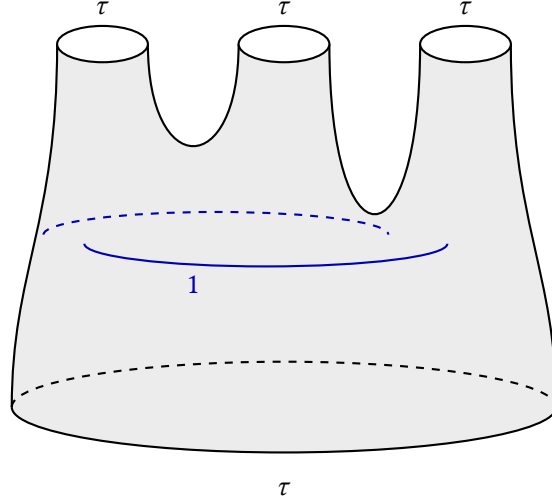
Generally when speaking about anyons, we group them into *theories*. A anyon theory is equipped with a (finite) set of charges $a, b, c, \dots$ including a *vacuum* charge I . Additionally we are provided with a *fusion rule* $a \times b = \sum_c N_{ab}^c c$ where N_{ab}^c denotes the multiplicity. Since it can be shown (should I?) that we cannot classify the vector space of two anyons a, b by itself, we need to parametrize this space over its *fusion sectors*, i.e. the outcomes c of the fusion rule.

Consider an anyon to be a point-like defect on the plane carrying a charge out of the set of labels. Since we need to reason about the properties of a pair of anyons using their fusion-outcome, we draw this as follows:



The thick dark ellipses here denote the fusion outcome of two charges. Note that the line here has an additional physical property in such that it is an *observable*. This has a nice interpretation in terms of bordisms:





Let V_c^{ab} be the space of 2 anyons of charges a, b fusing to c . Then the whole space V^{ab} splits accordingly:

$$V^{ab} = \bigoplus_{c \in I} V_c^{ab}.$$

Since we wanted to assign meaning to the *exchange* of particles, we define the exchange operator R^{ab} componentwise:

$$R^{ab} : V^{ab} \rightarrow V^{ba}, \quad \bigoplus_c V_c^{ab} \cong \bigoplus_c V_c^{ba}.$$

as well as an isomorphism on combinations of more than 3 particles

$$F^{abc} : V^{(ab)c} \cong V^{a(bc)}$$

6.1.1 Anyons on different topologies

Also we should do or have done isotopy before. Technically the discussion here also requires the completeness relations.

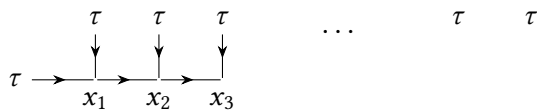
$$\begin{array}{c} a \\ \downarrow \\ \uparrow \\ \uparrow \end{array}
 \begin{array}{c} b \\ \downarrow \\ \uparrow \\ \uparrow \end{array}
 = \sum_{c, \mu} \sqrt{\frac{d_c}{d_a d_b}}
 \begin{array}{c} a \quad b \\ \searrow \quad \swarrow \\ \quad \downarrow \\ \quad \uparrow \\ \swarrow \quad \searrow \\ c \end{array}
 \quad (6.1)$$

6.2 1D LATTICE MODELS & THE GOLDEN CHAIN

What is a 2 + 1D critical point? This is relevant for the distinction between 1D and 2D models. Is a 2D model the same as a fusion surface model?

TODO: Write an outlook section here on work like [LDV24] [Eck25] etc. This seems like a direct generalization of our approach.

In [Fei+07], anyonic chains, specifically in the Fib model, are introduced as a 1D lattice of n τ anyons, defined as the fusion space thereof.



Here, the state is fully characterized by its intermediate indices $x_1 \dots x_{n-1}$, so we can choose a basis $|x_1 x_2 \dots x_{n-1}\rangle$.

Similar to the Heisenberg-chain¹, the golden chain introduces a Hamiltonian that assigns -1 to the interaction $\tau \otimes \tau = 1$ and 0 to the interaction $\tau \otimes \tau = \tau$.

We can decompose the whole space $\text{Hom}(\tau^{\otimes n+1}, \tau)$ into the local states

$$\text{Hom}(\tau^{\otimes n+1}, \tau) = \bigoplus_{\substack{x_1, x_2, \dots, x_{n-1} \\ \text{all admissible paths}}} \text{Hom}(\tau \otimes \tau, x_1) \otimes \text{Hom}(x_1 \otimes \tau, x_2) \otimes \dots \otimes \text{Hom}(x_{n-1} \otimes \tau, \tau) \quad (6.2)$$

with the admissibility condition that there are not adjacent 1 in the sequence of x_i s.

This is evaluated as a 3 local hamiltonian, as we can decompose the spin chain into local parts where two τ anyons are adjacent (check how they call it in the paper.).

$$\begin{array}{ccc} \begin{array}{c} \tau \quad \tau \\ \downarrow \quad \downarrow \\ x_{i-1} \quad x_i \quad x_{i+1} \end{array} & = \sum_{x_i} [F_{x_{i+1}}^{x_{i-1} \tau \tau}]_{x_i}^{x'_i} & \begin{array}{c} \tau \quad \tau \\ \swarrow \quad \searrow \\ x_{i-1} \quad x_{i+1} \end{array} \\ \in \text{Hom}(x_{i-1} \otimes \tau, x_i) & & \in \text{Hom}(x_{i-1} \otimes x'_i, x_{i+1}) \\ \otimes \text{Hom}(x_i \otimes \tau, x_{i+1}) & & \otimes \text{Hom}(\tau \otimes \tau, x'_i) \end{array}$$

Now we can apply the interaction based on the intermediate state² x'_i and apply another F -move back to the chain basis:

$$H_{[i, i+1]} : |x_{i-1} x_i x_{i+1}\rangle \rightarrow |x_{i-1} x'_i x_{i+1}\rangle := \sum_{x'_i \in \{1, \tau\}} [F_{x_{i+1}}^{x_{i-1} \tau \tau}]_{x_i}^{x'_i} E_{x'_i} [F_{x_{i+1}}^{x_{i-1} \tau \tau}]_{x'_i}^{x''_i}$$

In the golden chain, we set $E_1 = -1$ and $E_\tau = 0$. Then (6.2) reduces to

$$H_{[i, i+1]} = -[F_{x_{i+1}}^{x_{i-1} \tau \tau}]_1^{x_i} [F_{x_{i+1}}^{x_{i-1} \tau \tau}]_1^{x''_i}$$

and we can write this three local hamiltonian as a matrix

$$H_i = -J \begin{pmatrix} 1 & & & \\ & 0 & & \\ & & 0 & \\ & & \varphi^{-2} & \varphi^{-\frac{3}{2}} \\ & & \varphi^{-\frac{3}{2}} & \varphi^{-1} \end{pmatrix}$$

This method is reviewed and extended in [Tre+08], with longer ranger interaction. It will become apparent that the construction given in these papers is a manualization of the tensor network constructions we will build here. Generalizations to arbitrary models are given in [Gil+13] as well as [BG17]. Generalized spin chains and dualities using module categories are given in [Eck25].

The tensor network formalism is handled in [Pfe+10], [Pfe12] and [Aye17], and in terms of categories this is done in [Unf24] and [DH25].

If we want to write this as a tensor contraction, we first define a hamiltonian in a similar fashion to chapter 1.

This takes the form of an operator in $\text{Hom}(\tau \otimes \tau, \tau \otimes \tau)$ as

$$H_i = E_1 \begin{array}{c} \tau \quad \tau \\ \diagdown \quad \diagup \\ \bullet \\ | \\ \bullet \\ \diagup \quad \diagdown \\ \tau \quad \tau \end{array} 1 + E_\tau \begin{array}{c} \tau \quad \tau \\ \diagdown \quad \diagup \\ \bullet \\ | \\ \bullet \\ \diagup \quad \diagdown \\ \tau \quad \tau \end{array} \tau$$

¹Note that $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$ in the Heisenberg $\text{SU}(2)$ model.

²Technically we need a projector here read up again.

(Note this is allowed due to $\mathbb{C}$ -linearity of Fib.)

In this case we can view the golden chain approach chosen above as a manualization of the tensor network process.

We can also write H_i in its operator form matrix as

$$H_i \in \bigoplus_{a \in 1, \tau} \text{Hom}(\mathbb{C}, \mathbb{C})$$

as

$$H_i = -J \begin{array}{cc} & \begin{array}{cc} 1 & \tau \end{array} \\ \begin{array}{c} 1 \\ \tau \end{array} & \begin{pmatrix} E_1 & 0 \\ 0 & E_\tau \end{pmatrix} \end{array}$$

6.2.1 Topological Entanglement Entropy

For simple lattice states, [Sin+14] defines a bipartition of the lattice into anyons A and B . Then we consider the whole space under charge a_{tot} as

$$V_{a_{tot}} = \bigoplus_{N_{ab}^{a_{tot}}=1} (V_a^A \otimes V_b^B)$$

What i dont get here is why we are allowed to tensor both these sectors together without the space $V_{ab}^{a_{tot}}$ explicitly.

TODO delete all of this and introduce this over the Hom category.

Anyhow, the fundamental building blocks we are using are the vector spaces V_c^{ab} (where we refer to this as the fusion space from types a and b to c). This allows us to build the vector space of two anyonic types a, b as the sum over the fusion outcome:

$$V^{ab} = \bigoplus_{c \in \text{Irr}(C)} V_c^{ab}$$

We can directly introduce the R move (a braiding of the initial anyons) to obtain the isomorphic space V^{ba} as

$$R : \bigoplus_{c \in \text{Irr}(C)} V_c^{ab} \cong \bigoplus_{c \in \text{Irr}(C)} V_c^{ba}$$

We can also write the space $V_c^{ab} = \text{Hom}(a \otimes b, c)$ in the categorical language. Here we also need to realize that there is a difference between the isomorphism β and the action of R , which is really on the hom space $\text{Hom}(-, c)$. So this would be a yoneda perspective? Since we know R for all $a, b, c \in C$, we also know the action of R as an isomorphism. Well technically i believe that this allows us to determine for all simples c , and then semisimplicity extends this to all subjects (this would be nice to write down) and then the yoneda embedding is enough to know the whole space. Similarly, in our $\text{Rep}(G)$ categories (which i actually belive to be symmetric? Find this in etingof)

The algebra-category correspondence in the theory of anyons was first noted by [Kit06, App E]

6.3 TOPOLOGICAL QUANTUM COMPUTATION

We want to build a quantum computer that acts purely on the topological data provided by the anyon theories, or as named by Freedman [Fre98], a *Quantum Field Computer*. To do this we need 3 things:

A ground state that has high enough dimension to represent a quantum bit, i.e. is $\cong \mathbb{C}^2$.

A option to time evolve the system by application of unitaries.

An option to read out the information of the evolved state (*measurement*).

Computing with nonabelian anyons requires a universal gate set built from representations of the *braid-group*:

Definition 6.3.1 (Braid Group (algebraic)).

$$Br(\mathbb{R}^2) = \left\langle \begin{array}{l} \sigma_i \sigma_j = \sigma_j \sigma_i, \quad |i - j| \geq 2 \\ \sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}, \quad i \in [1, n - 2] \end{array} \right\rangle$$

and also the *pure-braid-group*:

By the construction of anyon chains in Section 6.1, we know that the space of 3 anyons fusing to τ is in fact 2-dimensional. This motivates a choice of qubits as the following:

$$|0\rangle = \begin{array}{c} \text{Diagram: An oval containing three dots labeled } \tau. \text{ A smaller oval inside contains two dots labeled } \tau. \text{ The label } 1 \text{ is at the bottom right.} \end{array} \quad |1\rangle = \begin{array}{c} \text{Diagram: An oval containing three dots labeled } \tau. \text{ A smaller oval inside contains two dots labeled } \tau. \text{ The label } \tau \text{ is at the bottom right.} \end{array} \quad |NC\rangle = \begin{array}{c} \text{Diagram: An oval containing three dots labeled } \tau. \text{ A smaller oval inside contains two dots labeled } \tau. \text{ The label } 1 \text{ is at the bottom right.} \end{array}$$

and we say that $\mathbb{C}^2$ is *embedded* in $V^{\tau\tau\tau}$. We could choose a different presentations, where each anyon is drawn from the vacuum as

TODO!

Then we can build multi-qubit circuits by simply stacking these diagrams (inside their own fusion channel). This is what is called a *sparse-encoding*. A dense encoding would correspond to just choosing the smallest space of appropriate dimension that includes one of the $(\mathbb{C}^2)^{\otimes d}$.

The solovay-kitaev theorem guarantees a polylog time and space algorithm for approximating any gate in $SU(d)$ to a desired accuracy. This, and similar procedures are realized by the functor *compile*. A *topological quantum compiler* essentially only translates the qubits to their Hom space equivalences and translates the qudit-gates to their braid group representations. There are of course special compilation methods, the one we cover here takes first a decomposition of normal gates into CNOTs and then applies injection weaving, so we only need to lower single qubit gates to braids.

$$\begin{array}{ccc} (\mathbb{C}^2)^{\otimes n} & \xrightarrow{\text{compile}} & \text{Hom}(\tau^{2n+2}, a) \\ \downarrow U & \text{Commutates up to} & \downarrow \rho(b) \\ & \text{given } \epsilon = |U - \rho(b)| & \\ (\mathbb{C}^2)^{\otimes n} & \xrightarrow{\text{compile}} & \text{Hom}(\tau^{2n+2}, a) \end{array}$$

Because this square commutes and *compile* runs in polynomial time, we can in fact assert that the *QFC* can simulate *BQP*.

However, compiling qudits for braids is a *hard* (computationally expensive) and slow process, as the dimension of the embedding is usually a constant factor > 1 larger than the dimension of the desired qudit space.

This process (and these do depend on the topology of the ground state) and the individual gates are given in [HG24], [GHM24].

The work of Simon et al. [Sim+06] provides an alternative using the pure braid group. Remember that although we can represent any weave generator as

$$T_{ij} = \sigma_i \sigma_{i+1} \cdots \sigma_{j-1} \sigma_j^2 \sigma_{j-1} \cdots \sigma_{i+1} \sigma_i$$

a weave is effectively realized by moving only one of the particles (strands) around $n - 1$ fixed strands. As it turns out, the weave representations are also dense in $SU((2))$. (cite). This can also solve our problem of effectively compiling controlled gates. Remember the presentation we chose for our qubits: The $|0\rangle$ qubit has an intermediate channel of 1 and the $|1\rangle$ qubit has an intermediate channel of τ . We also know that braiding a 1-type anyon around any other anyon should have no effect. By our construction of fusion we know that the tau anyons in the 1 channel can only produce 1 when fused (i.e. $\tau \otimes \tau = 1$). Should we have a superposition of qubits $\alpha |0\rangle + \beta |1\rangle$ this means we produce a composite anyon of type 0 with probability α^2 and a composite anyon of type τ with probability β^2 . Now if we find a weave realizing our controlled gate on the lower qubit,

we will have successfully synthesized a controlled gate. For this, we only need to *inject* the composite anyon in the lower qubit *without* affecting the state of our system. After fusing in our top qubit, we effectively end up with a braiding problem in PBr_n , however since we only want the τ part to have an effect we can reduce this to Br_n . Practically, this means that we need to find a traditional braid $X \in Br_3$ such that the representation of

or doing so invertibly

$$\sigma_3$$

We can use *weaves* to build *injection gates* which in turn realize control gates. Then, by a universality argument of [FLW00] we can approximate any other multiqubit gate, as single qubit gates can be computed in little time after an algorithms³⁴ of [KBS13] based on number theoretic methods (which are already mentioned in [FW07]).

Rewrite this.

Specifically, there is a decomposition of an arbitrary gate in $(\mathbb{C}^2)^{\otimes n}$ into a universal set of *CNOT* and $\mathbb{C}^2$ unitaries. All that remains to do is to reexpress a traditional working gate set in the *Fib* basis of sets.

However we must also ensure that the decomposition into CNOT does not produce nonadjacent control-target pairs, as we can not freely cross the braids over another qubit. This can however also be a problem for standard qubit architectures, which is why we can decompose every unitary circuit into its *linear nearest neighbour* (LNN) form [KMS07; SWD11]. Such decompositions are already implemented in languages such as Qiskit⁵ [Jav+24].

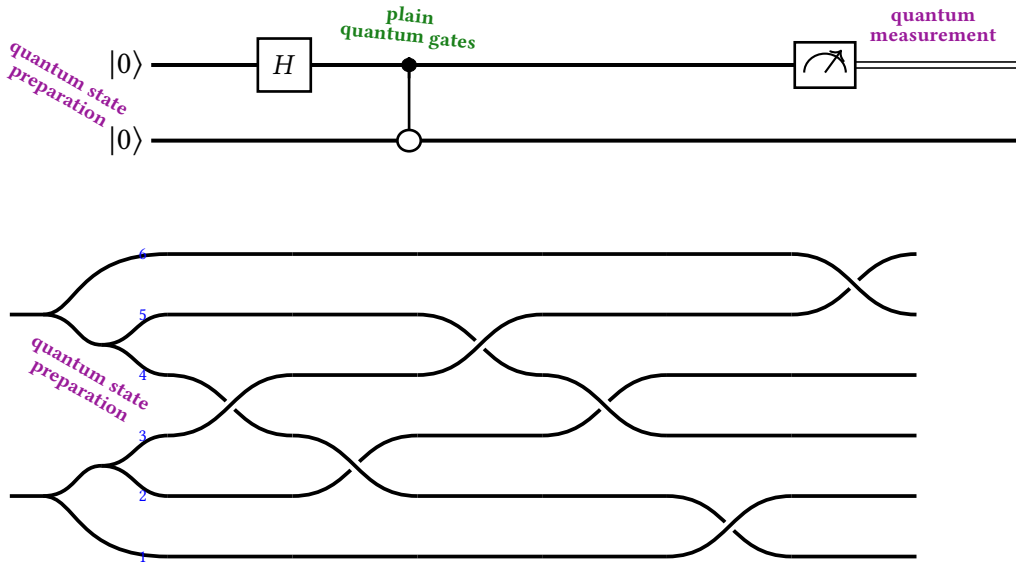


Figure 6.1: A standard 2-qubit quantum circuit and its lowering to a braid-circuit with injection weaves (missing yet).

6.3.1 Application: Evaluating the Jones Polynomial

³We provide an implementation of this process in Haskell [Rie25]

⁴Although we cannot reach the claimed bound of ϵ but instead reach $\sqrt{\epsilon}$

⁵Qiskit documentation: <https://quantum.cloud.ibm.com/docs/en/api/qiskit>

A

APPENDIX

A.1 SOME GROUP THEORY

Definition A.1.1. A *group* $(G, \cdot)$ consists of a (possibly infinite) set of elements and a binary op $\cdot : G \times G \rightarrow G$ such that

1. $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
2. There exists an $\text{id}_G \in G$ s.t. $\text{id}_G \cdot a = a \cdot \text{id}_G = a$
3. Each $a \in G$ is invertible with $a \cdot a^{-1} = a^{-1}a = \text{id}_G$

Definition A.1.2. A group $(G, \cdot)$ is called *abelian*, if for all $a, b \in G$ it holds that $a \cdot b = b \cdot a$, its contrapose is called a *non-abelian* group.

Recall that we say a group is *generated* by a set of elements if we can write any other element as the concatenation of the generating set. Here by example:

Definition A.1.3. The *symmetric group* of n points S_n is given by

$$S_n = \left\langle s_1, \dots, s_{n-1} \left| \begin{array}{l} s_i^2 = \text{id} \\ s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1} \\ s_i s_j = s_j s_i, |i - j| \geq 2 \end{array} \right. \right\rangle. \quad (\text{A.1})$$

Definition A.1.4 (Representation theory of the symmetric group). For $n \geq 2$ the symmetric group S_n is equipped with two one-dimensional representations, the *trivial representation* where every generator is equal to the identity, and the *sign representation* which equips each generator with its relative permutation parity.

Definition A.1.5. The *General Linear Group* $\text{GL}(n)$ of degree n is the set of $n \times n$ matrices that are invertible, and its operation $\cdot$ is that of matrix multiplication.

A.2 SOME ALGEBRAIC TOPOLOGY

Let spaces X, Y be topological in the following, and let all functions be continuous unless stated otherwise.

Definition A.2.1. A *homotopy* between two continuous functions $f, g : X \rightarrow Y$ is a continuous function

$$H : X \times [0, 1] \rightarrow Y \text{ such that } H(x, 0) = f(x) \text{ and } H(x, 1) = g(x) \quad (\text{A.2})$$

Definition A.2.2. A *loop* with basepoint x_0 on X is a function

$$\gamma : [0, 1] \rightarrow X \text{ such that } \gamma(0) = \gamma(1) = x_0 \quad (\text{A.3})$$

A.3 PROOFS FOR CHAPTER 3

This is the proof for statement Equation (3.8) on page 11.

Lemma A.3.1. *The XXZ Hamiltonian admits a symmetry under $U(\theta) = e^{i\theta S_{tot}^z}$ and diagonalizing with S_{tot}^z suffices.*

Proof. Actually should we just cite a proof or is this of interest? [Has24, § 4.2]. Recall that when two operators commute, we can *simultaneously diagonalize* them [HJ12, Theorem 1.3.12].

$$[H, S_{tot}^z] = \sum_i^N [\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y + \Delta \sigma_i^z \sigma_i^z, \sum_{j=1}^N \sigma_j^z]$$

and since operators on different sites commute, only $j = i$ and $j = i + 1$ contribute. So we ought to write

$$\sum_i^N [\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y + \Delta \sigma_i^z \sigma_i^z, \sigma_i^z + \sigma_{i+1}^z]$$

and now using the identity $[AB, C] = A[B, C] + [A, C]B$:

$$[\sigma_i^x \sigma_{i+1}^x, \sigma_i^z + \sigma_{i+1}^z] = -2i(\sigma_i^y \sigma_{i+1}^x + \sigma_i^x \sigma_{i+1}^y)$$

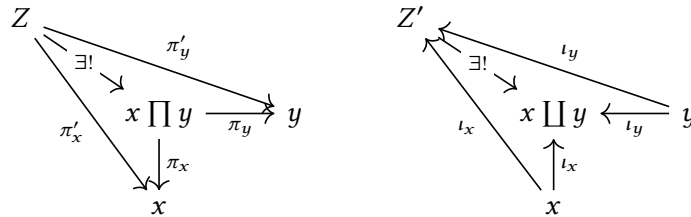
$$[\sigma_i^y \sigma_{i+1}^y, \sigma_i^z + \sigma_{i+1}^z] = 2i(\sigma_i^x \sigma_{i+1}^y + \sigma_i^y \sigma_{i+1}^x)$$

which cancel out. Furthermore, the z spins commute as well. Thus the whole sum is 0 and we are done. $\square$

A.4 SUPPLEMENTARY CATEGORICAL CONSTRUCTIONS FOR CHAPTER 5

Note on universal properties: When dealing with categories, we are fundamentally dealing with abstractions that behave alike to a known construction. These behaviours are made precise by the use of a *universal property*. We illustrate this with the definition of a (co)product:

Definition A.4.1. Let C be a category and $x, y \in ob(C)$. Their *product* is an object $x \amalg y$ in C with morphisms $\pi_x : x \amalg y \rightarrow x$ and $\pi_y : x \amalg y \rightarrow y$. Their *coproduct* is defined as the object $x \sqcup y$ together with the maps $\iota_x : x \rightarrow x \sqcup y$ and $\iota_y : y \rightarrow x \sqcup y$. Both of these definitions are universal with their properties.



This should all go in the anyon chapter? or later on into categories. Lets show how to build the splitting of a fusion tree

$$\text{Hom}(a \otimes b \otimes d, e) = \bigoplus_c \text{Hom}(a \otimes b, c) \otimes \text{Hom}(c \otimes d, e)$$

First we ought to give an mapping

$$\Theta_c : \text{Hom}(a \otimes b, c) \otimes_k \text{Hom}(c \otimes d, e) \rightarrow \text{Hom}(a \otimes b \otimes d, e)$$

by

$$\mu \otimes_k v \mapsto (\mu \otimes \text{id}_d) \circ v$$

A.4.1 TQFT

Hmm, where does this go?

A TQFT is defined by a finite set of labels L , and a rule that assigns each compact cpa-space a vector space and each map between them (a cobordism) a linea map between spaces. A TQFT is thus a functor between the category $(Bord, \sqcup, \emptyset) \rightarrow Vect$.

We can interpret this as assigning each ??? bordism its space of states under the functor and each cobordism its space of operators.

A.5 OVERVIEW OF EXISTING IMPLEMENTATIONS

Table A.1: Overview of tensor network libraries and their support for symmetries and features relevant toward symmetry. If a specific symmetry is given, the software is restricted to that symmetry. A previous list of existing implementations in 2018 is also given in [Hub18]. Note that implementations such as the original MATLAB implementation by Singh, Pfeifer, and Vidal [SPV10] are not publicly available or named.

Metadata				Features							Software & Publication
Library/Backend	Year	Language	License	Abelian	Non-Abelian	Categorical	Product Symmetries	Fermionic Number	GPU	Autodiff	
MPToolkit	(L) ¹	2002	C++	GPL-3.0	✓	✓	✗	✓	?	✗	[26b; MG02]
QSpace	(L)	2012	Matlab	Apache-2.0	✓	✓	✗	?	?	?	[Wei24; Wei12a]
CheMPS2	(L)	2013	C++	GPL-2.0	✓	✓	Ferm	✗	✓	?	[Wou+14]
Uni10	(L)	2015	Python, C++	?	?	?	?	?	?	?	[KHC15]
SyTen	(L)	2017	C++	✗ ²	?	?	?	?	?	?	[Hub+; Hub17]
SymLibrary	(R)	2019	Matlab	MIT	✓	✓	$SU(2)_k$	✓	?	?	[McM19; MSB20]
TensorNetwork	(L)	2019	Python	?	✓	✗	✗	✗	?	?	[Rob+19]
TenPy	(L)	2019	Python	Apache-2.0	✓	✗	✗	?	?	?	[Hau+24]
ITensor	(L)	2022	Julia	Apache-2.0	✓	✓	?	?	?	?	[FWS22]
Block2	(L)	2023	C++	GPL-3.0	✓	$SU(2)$	✗	✗	?	?	[26a; Zha+23]
ChemTensor	(L)	2023	C	Apache-2.0	✓	$SU(2)$	✗	✗	?	✗	[Men26]
grassmannntn	(R)	2023	Python	Apache-2.0			sVect ³	?	✓	?	[Yos26]
QuantumTea	(L)	2024	Python	?	✓	?	?	?	?	?	[Bac+26]
Cyten ²⁴	(L)(B)	2024	Python & C++	Apache-2.0	✓	✓	✓	✓	✓	✗	[26c; Unf24]
YASTN		2025	Python	?	✓	✗	✗	✓	?	✓	[Ram+25]
TensorKit.jl ⁵	(L)	2025	Julia	MIT	✓	✓	✓	✓	?	✓	[DH25]
Cytnx ⁴		2024	C++	?	✓	✗	✗	✓	?	✓	[Wu+25]
TeMFpy ⁸	(B)	2025	Python	MIT	✗	✗	✗	✗	✓	✗	[HS25]
abeliantensors		2026	Python	MIT	$U(1), \mathbb{Z}_n$	✗	✗	✗	✗	✗	[Hau26]
symmray ⁶	(B)	2025	Python	?	✓	✗	sVect	✓	✓	?	[Gao+25; Gra26b]
Telum.jl		2026	Julia	?	✓	✓	Ferm	?	✓	?	[Lee26]
Su2tn	(R)	2025	Python	None	✗	$SU(2)$	✗	✗	7	?	[Fis25b; Fis25a]

¹(L) indicates that this is a standalone Library, (B) indicates that this is a backend for another library and (R) indicates that this is a research script and likely not maintained.

²Cyten originated as a branch of TenPy and was later

split into its own repository [Tenpy pull 309].

³This supports a variant called *Grassman Tensor Networks*.

⁴Ideas from Cytnx were eventually adapted into Cyten.

⁵Part of the QuantumKit.jl ecosystem.

⁶symmray functions as a backed for the tensor network software quimb [Gra18; Gra26a]

⁷Syten is a closed source program available among collaborations.

⁸TeMFpy is built on top of Tenpy.

BIBLIOGRAPHY

- [26a] *Block-Hczhai/Block2-Preview*. block-hczhai, Sept. 17, 2026. URL: <https://github.com/block-hczhai/block2-preview> (visited on 09/17/2026) (cit. on p. 41).
- [26b] *Mptoolkit/Mptoolkit*. mptoolkit, Sept. 7, 2026. URL: <https://github.com/mptoolkit/mptoolkit> (visited on 09/17/2026) (cit. on p. 41).
- [26c] *Tenpy/Cyten*. Tensor Network Python Package, Sept. 16, 2026. URL: <https://github.com/tenpy/cyten> (visited on 09/17/2026) (cit. on p. 41).
- [Abr12] Samson Abramsky. *No-Cloning In Categorical Quantum Mechanics*. Mar. 19, 2012. DOI: [10.48550/arXiv.0910.2401](https://doi.org/10.48550/arXiv.0910.2401). arXiv: [0910.2401](https://arxiv.org/abs/0910.2401) [quant-ph]. (Visited on 07/26/2026). Pre-published (cit. on p. 4).
- [AK23] Fatimah Rita Ahmadi and Aleks Kissinger. “The ZX-Calculus as a Language for Topological Quantum Computation”. In: *Journal of Physics A: Mathematical and Theoretical* 56.41 (Oct. 13, 2023), p. 415301. DOI: [10.1088/1751-8121/acef7e](https://doi.org/10.1088/1751-8121/acef7e). arXiv: [2211.03855](https://arxiv.org/abs/2211.03855) [quant-ph]. (Visited on 11/21/2024) (cit. on p. 4).
- [Aye17] Babatunde Moses Ayeni. “Studies of Braided Non-Abelian Anyons Using Anyonic Tensor Networks”. PhD thesis. Macquarie University, Dec. 3, 2017. DOI: [10.25949/19435697.v1](https://doi.org/10.25949/19435697.v1). arXiv: [1708.06476](https://arxiv.org/abs/1708.06476) [quant-ph]. (Visited on 06/28/2026) (cit. on pp. 19, 35).
- [Bac+26] Flavio Baccari et al. *Quantum TEA: Qtealeaves*. Version 1.12.5. Zenodo, May 2026. DOI: [10.5281/zenodo.20109357](https://doi.org/10.5281/zenodo.20109357) (cit. on p. 41).
- [Bai17] James Bailie. *Semisimple Categories*. 2017. URL: <https://jameshbailie.github.io/files/papers/2017-06-26-Semisimple-categories.pdf> (visited on 07/08/2026) (cit. on p. 25).
- [BG17] Matthew Buican and Andrey Gromov. “Anyonic Chains, Topological Defects, and Conformal Field Theory”. In: *Communications in Mathematical Physics* 356.3 (Dec. 1, 2017), pp. 1017–1056. DOI: [10.1007/s00220-017-2995-6](https://doi.org/10.1007/s00220-017-2995-6). arXiv: [1701.02800v2](https://arxiv.org/abs/1701.02800v2) [hep-th]. (Visited on 06/28/2026) (cit. on pp. 4, 35).
- [BKP17] Parsa Bonderson, Christina Knapp, and Kaushal Patel. “Anyonic Entanglement and Topological Entanglement Entropy”. In: *Annals of Physics* 385 (Oct. 1, 2017), pp. 399–468. DOI: [10.1016/j.aop.2017.07.018](https://doi.org/10.1016/j.aop.2017.07.018). (Visited on 07/05/2026) (cit. on p. 4).
- [Bon07] Parsa Hassan Bonderson. “Non-Abelian Anyons and Interferometry”. California Institute of Technology, June 4, 2007. DOI: [10.7907/5NDZ-W890](https://doi.org/10.7907/5NDZ-W890). (Visited on 07/26/2026) (cit. on p. 28).
- [BSS08] Parsa Bonderson, Kirill Shtengel, and J. K. Slingerland. “Interferometry of Non-Abelian Anyons”. In: *Annals of Physics* 323.11 (Nov. 1, 2008), pp. 2709–2755. DOI: [10.1016/j.aop.2008.01.012](https://doi.org/10.1016/j.aop.2008.01.012). (Visited on 07/26/2026) (cit. on pp. 4, 28).
- [Bur16] Simon Burton. *A Short Guide to Anyons and Modular Functors*. Oct. 17, 2016. DOI: [10.48550/arXiv.1610.05384](https://doi.org/10.48550/arXiv.1610.05384). arXiv: [1610.05384](https://arxiv.org/abs/1610.05384) [quant-ph]. (Visited on 07/15/2026). Pre-published (cit. on pp. 4, 18, 25).
- [CB08] Gregory M. Crosswhite and Dave Bacon. “Finite Automata for Caching in Matrix Product Algorithms”. In: *Physical Review A* 78.1 (July 29, 2008), p. 012356. DOI: [10.1103/PhysRevA.78.012356](https://doi.org/10.1103/PhysRevA.78.012356). (Visited on 05/18/2026) (cit. on p. 7).

- [Cir+17] J. I. Cirac et al. “Matrix Product Density Operators: Renormalization Fixed Points and Boundary Theories”. In: *Annals of Physics* 378 (Mar. 1, 2017), pp. 100–149. DOI: [10.1016/j.aop.2016.12.030](https://doi.org/10.1016/j.aop.2016.12.030). arXiv: [1606.00608v4](https://arxiv.org/abs/1606.00608v4) [quant-ph] (cit. on p. 15).
- [Del02] P. Deligne. “Categories Tensorielles”. In: *Moscow Mathematical Journal* 2.2 (2002), pp. 227–248. DOI: [10.17323/1609-4514-2002-2-2-227-248](https://doi.org/10.17323/1609-4514-2002-2-2-227-248). (Visited on 09/23/2026) (cit. on p. 19).
- [Del90] Pierre Deligne. “Catégories Tannakiennes”. In: *The Grothendieck Festschrift, Volume II*. Ed. by Pierre Cartier et al. Vol. 87. Progress in Mathematics. Boston, MA: Birkhäuser, 1990, pp. 111–195 (cit. on p. 19).
- [DH25] Lukas Devos and Jutho Haegeman. *TensorKit.jl: A Julia Package for Large-Scale Tensor Computations, with a Hint of Category Theory*. Version 1. 2025. DOI: [10.48550/arxiv.2508.10076](https://doi.org/10.48550/arxiv.2508.10076). (Visited on 06/01/2026). Pre-published (cit. on pp. 2, 3, 19, 30, 35, 41).
- [DN05] Christopher M. Dawson and Michael A. Nielsen. *The Solovay-Kitaev Algorithm*. Aug. 23, 2005. DOI: [10.48550/arXiv.quant-ph/0505030](https://doi.org/10.48550/arXiv.quant-ph/0505030). arXiv: [quant-ph/0505030](https://arxiv.org/abs/quant-ph/0505030). (Visited on 07/15/2026). Pre-published (cit. on p. 4).
- [DT24] Clement Delcamp and Apoorv Tiwari. “Higher Categorical Symmetries and Gauging in Two-Dimensional Spin Systems”. In: *SciPost Physics* 16.4 (Apr. 23, 2024), p. 110. DOI: [10.21468/SciPostPhys.16.4.110](https://doi.org/10.21468/SciPostPhys.16.4.110). arXiv: [2301.01259](https://arxiv.org/abs/2301.01259) [hep-th]. (Visited on 06/08/2026) (cit. on p. 5).
- [Eck25] Luisa Eck. “Quantum Lattice Models from Fusion Categories”. PhD thesis. University of Oxford, 2025. DOI: [10.5287/ora-zozm87o5w](https://doi.org/10.5287/ora-zozm87o5w). (Visited on 06/28/2026) (cit. on pp. 34, 35).
- [ENO17] Pavel Etingof, Dmitri Nikshych, and Viktor Ostrik. *On Fusion Categories*. Apr. 28, 2017. DOI: [10.48550/arXiv.math/0203060](https://doi.org/10.48550/arXiv.math/0203060). arXiv: [math/0203060](https://arxiv.org/abs/math/0203060). (Visited on 09/23/2026). Pre-published (cit. on p. 19).
- [FB62] Edward Fadell and James Van Buskirk. “The Braid Groups of E_2 and S^2 ”. In: *Duke Mathematical Journal* 29.2 (June 1962), pp. 243–257. DOI: [10.1215/S0012-7094-62-02925-3](https://doi.org/10.1215/S0012-7094-62-02925-3). (Visited on 07/28/2026) (cit. on p. 18).
- [Fei+07] Adrian Feiguin et al. “Interacting Anyons in Topological Quantum Liquids: The Golden Chain”. In: *Physical Review Letters* 98.16 (Apr. 20, 2007), p. 160409. DOI: [10.1103/PhysRevLett.98.160409](https://doi.org/10.1103/PhysRevLett.98.160409). (Visited on 06/28/2026) (cit. on pp. 4, 34).
- [Fis25a] Victor Fischer. “Tensor Network Methods with Non-Abelian Symmetries”. MA thesis. Munich: Technische Universität München, 2025 (cit. on p. 41).
- [Fis25b] Victor Fischer. *Victorfischer2000/Su2tn*. Feb. 4, 2025. URL: <https://github.com/victorfischer2000/su2tn> (visited on 08/17/2026) (cit. on p. 41).
- [FLW00] Michael Freedman, Michael Larsen, and Zhenghan Wang. *A Modular Functor Which Is Universal for Quantum Computation*. Feb. 1, 2000. DOI: [10.48550/arXiv.quant-ph/0001108](https://doi.org/10.48550/arXiv.quant-ph/0001108). arXiv: [quant-ph/0001108](https://arxiv.org/abs/quant-ph/0001108). (Visited on 08/13/2026). Pre-published (cit. on p. 38).
- [FNW92] M. Fannes, B. Nachtergaele, and R. F. Werner. “Finitely Correlated States on Quantum Spin Chains”. In: *Communications in Mathematical Physics* 144.3 (Mar. 1, 1992), pp. 443–490. DOI: [10.1007/BF02099178](https://doi.org/10.1007/BF02099178). (Visited on 07/15/2026) (cit. on p. 3).
- [Fre98] Michael H. Freedman. “P/NP, and the Quantum Field Computer”. In: *Proceedings of the National Academy of Sciences* 95.1 (Jan. 6, 1998), pp. 98–101. DOI: [10.1073/pnas.95.1.98](https://doi.org/10.1073/pnas.95.1.98). (Visited on 07/25/2026) (cit. on pp. 4, 36).
- [FW07] Michael H. Freedman and Zhenghan Wang. “Large Fourier Transforms Never Exactly Realized by Braiding Conformal Blocks”. In: *Physical Review A* 75.3 (Mar. 15, 2007), p. 032322. DOI: [10.1103/PhysRevA.75.032322](https://doi.org/10.1103/PhysRevA.75.032322). arXiv: [cond-mat/0609411](https://arxiv.org/abs/cond-mat/0609411). (Visited on 08/03/2026) (cit. on p. 38).
- [FWS22] Matthew Fishman, Steven R. White, and E. Miles Stoudenmire. “The ITensor Software Library for Tensor Network Calculations”. In: *SciPost Phys. Codebases* (2022), p. 4. DOI: [10.21468/SciPostPhysCodeb.4](https://doi.org/10.21468/SciPostPhysCodeb.4) (cit. on p. 41).

- [Gao+25] Yang Gao et al. “Fermionic Tensor Network Contraction for Arbitrary Geometries”. In: *Physical Review Research* 7.2 (May 27, 2025), p. 023193. DOI: [10.1103/PhysRevResearch.7.023193](https://doi.org/10.1103/PhysRevResearch.7.023193). (Visited on 05/04/2026) (cit. on p. 41).
- [GHM24] Lachezar S. Georgiev, Ludmil Hadjiivanov, and Grigori Matein. *Diagonal Coset Approach to Topological Quantum Computation with Fibonacci Anyons*. Apr. 3, 2024. DOI: [10.48550/arXiv.2404.01779](https://doi.org/10.48550/arXiv.2404.01779). arXiv: [2404.01779](https://arxiv.org/abs/2404.01779) [quant-ph]. (Visited on 08/13/2026). Pre-published (cit. on p. 37).
- [Gil+13] C. Gils et al. “Anyonic Quantum Spin Chains: Spin-1 Generalizations and Topological Stability”. In: *Physical Review B* 87.23 (June 17, 2013), p. 235120. DOI: [10.1103/PhysRevB.87.235120](https://doi.org/10.1103/PhysRevB.87.235120). (Visited on 06/28/2026) (cit. on pp. 4, 35).
- [Gra18] Johnnie Gray. “Quimb: A Python Library for Quantum Information and Many-Body Calculations”. In: *Journal of Open Source Software* 3.29 (2018), p. 819. DOI: [10.21105/joss.00819](https://doi.org/10.21105/joss.00819) (cit. on p. 41).
- [Gra26a] Johnnie Gray. *Jcmgray/Quimb*. Sept. 17, 2026. URL: <https://github.com/jcmgray/quimb> (visited on 09/17/2026) (cit. on p. 41).
- [Gra26b] Johnnie Gray. *Jcmgray/Symmray*. June 21, 2026. URL: <https://github.com/jcmgray/symmray> (visited on 06/30/2026) (cit. on p. 41).
- [Has24] Fabian Hassler. *Show_document*. Lecture Notes. 2024. URL: https://www.quantuminfo.physik.rwth-aachen.de/global/show_document.asp?id=aaaaaaaacmgbnkz&utm_source=chatgpt.com (visited on 07/31/2026) (cit. on p. 40).
- [Hau+24] Johannes Hauschild et al. “Tensor Network Python (TeNPy) Version 1”. In: *SciPost Phys. Codebases* (2024), p. 41. DOI: [10.21468/SciPostPhysCodeb.41](https://doi.org/10.21468/SciPostPhysCodeb.41) (cit. on p. 41).
- [Hau26] Markus Hauru. *Mhauru/Abeliantensors*. May 16, 2026. URL: <https://github.com/mhauru/abeliantensors> (visited on 06/30/2026) (cit. on p. 41).
- [HG24] Ludmil Hadjiivanov and Lachezar S. Georgiev. “Braiding Fibonacci Anyons”. In: *Journal of High Energy Physics* 2024.8 (Aug. 12, 2024), p. 84. DOI: [10.1007/JHEP08\(2024\)084](https://doi.org/10.1007/JHEP08(2024)084). arXiv: [2404.01778](https://arxiv.org/abs/2404.01778) [hep-th]. (Visited on 08/13/2026) (cit. on p. 37).
- [HJ12] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. 2nd ed. Cambridge: Cambridge University Press, 2012. ISBN: 978-1-139-02041-1. DOI: [10.1017/CBO9781139020411](https://doi.org/10.1017/CBO9781139020411). (Visited on 06/30/2026) (cit. on pp. 10, 40).
- [HS25] Simon H. Hille and Attila Szabó. *TeMFpy: A Python Library for Converting Fermionic Mean-Field States into Tensor Networks*. 2025. arXiv: [2510.05227](https://arxiv.org/abs/2510.05227) [cond-mat.str-el] (cit. on p. 41).
- [Hub+] Claudius Hubig et al. *The SYTEN Toolkit*. URL: <https://syten.eu> (cit. on p. 41).
- [Hub17] Claudius Hubig. “Symmetry-Protected Tensor Networks”. PhD thesis. LMU München, 2017. URL: <https://edoc.ub.uni-muenchen.de/21348/> (cit. on p. 41).
- [Hub18] Claudius Hubig. “Global Symmetries in Tensor Networks”. 2018. URL: https://nextcloud.tfk.ph.tum.de/etn/wp-content/uploads/2018/11/201811_dresden.pdf (visited on 09/16/2026) (cit. on p. 41).
- [IR19] Alberto Ibort and Miguel A. Rodriguez. *An Introduction to Groups, Groupoids and Their Representations*. Boca Raton: CRC Press, Oct. 28, 2019. 362 pp. ISBN: 978-1-315-23294-2. DOI: [10.1201/b22019](https://doi.org/10.1201/b22019) (cit. on p. 25).
- [Jav+24] Ali Javadi-Abhari et al. *Quantum Computing with Qiskit*. 2024. DOI: [10.48550/arXiv.2405.08810](https://doi.org/10.48550/arXiv.2405.08810). arXiv: [2405.08810](https://arxiv.org/abs/2405.08810) [quant-ph] (cit. on p. 38).
- [JS91] André Joyal and Ross Street. “The Geometry of Tensor Calculus, I”. In: *Advances in Mathematics* 88.1 (July 1, 1991), pp. 55–112. DOI: [10.1016/0001-8708\(91\)90003-P](https://doi.org/10.1016/0001-8708(91)90003-P). (Visited on 09/23/2026) (cit. on pp. 19, 22).
- [KBS13] Vadym Kliuchnikov, Alex Bocharov, and Krysta M. Svore. *Asymptotically Optimal Topological Quantum Compiling*. arXiv.org. Oct. 15, 2013. DOI: [10.1103/PhysRevLett.112.140504](https://doi.org/10.1103/PhysRevLett.112.140504). (Visited on 08/03/2026) (cit. on p. 38).

- [KHC15] Ying-Jer Kao, Yun-Da Hsieh, and Pochung Chen. “Uni10: An Open-Source Library for Tensor Network Algorithms”. In: *Journal of Physics: Conference Series* 640 (Sept. 28, 2015), p. 012040. doi: [10.1088/1742-6596/640/1/012040](https://doi.org/10.1088/1742-6596/640/1/012040). (Visited on 09/16/2026) (cit. on p. 41).
- [Kir+23] Nico Kirchner et al. “Numerical Simulation of Non-Abelian Anyons”. In: *Physical Review B* 107.19 (May 15, 2023), p. 195129. doi: [10.1103/PhysRevB.107.195129](https://doi.org/10.1103/PhysRevB.107.195129). arXiv: [2206.14730](https://arxiv.org/abs/2206.14730) [cond-mat.str-el]. (Visited on 07/16/2026) (cit. on p. 4).
- [Kit03] A. Yu Kitaev. “Fault-Tolerant Quantum Computation by Anyons”. In: *Annals of Physics* 303.1 (Jan. 2003), pp. 2–30. doi: [10.1016/S0003-4916\(02\)00018-0](https://doi.org/10.1016/S0003-4916(02)00018-0). arXiv: [quant-ph/9707021](https://arxiv.org/abs/quant-ph/9707021). (Visited on 07/17/2026) (cit. on p. 4).
- [Kit06] Alexei Kitaev. “Anyons in an Exactly Solved Model and Beyond”. In: *Annals of Physics* 321.1 (Jan. 2006), pp. 2–111. doi: [10.1016/j.aop.2005.10.005](https://doi.org/10.1016/j.aop.2005.10.005). arXiv: [cond-mat/0506438v3](https://arxiv.org/abs/cond-mat/0506438v3). (Visited on 07/14/2026) (cit. on pp. 4, 19, 28, 36).
- [Kit97] A Yu Kitaev. “Quantum Computations: Algorithms and Error Correction”. In: *Russian Mathematical Surveys* 52.6 (Dec. 31, 1997), pp. 1191–1249. doi: [10.1070/RM1997v052n06ABEH002155](https://doi.org/10.1070/RM1997v052n06ABEH002155). (Visited on 07/26/2026) (cit. on p. 4).
- [KMS07] Samuel A. Kutin, David Petrie Moulton, and Lawren M. Smithline. *Computation at a Distance*. Jan. 26, 2007. doi: [10.48550/arXiv.quant-ph/0701194](https://doi.org/10.48550/arXiv.quant-ph/0701194). arXiv: [quant-ph/0701194](https://arxiv.org/abs/quant-ph/0701194). (Visited on 09/10/2026). Pre-published (cit. on p. 38).
- [Kon+20] Liang Kong et al. “Algebraic Higher Symmetry and Categorical Symmetry: A Holographic and Entanglement View of Symmetry”. In: *Physical Review Research* 2.4 (Oct. 15, 2020), p. 043086. doi: [10.1103/PhysRevResearch.2.043086](https://doi.org/10.1103/PhysRevResearch.2.043086). (Visited on 07/05/2026) (cit. on p. 5).
- [KZ22] Liang Kong and Zhi-Hao Zhang. *An Invitation to Topological Orders and Category Theory*. May 31, 2022. doi: [10.48550/arXiv.2205.05565](https://doi.org/10.48550/arXiv.2205.05565). arXiv: [2205.05565](https://arxiv.org/abs/2205.05565) [cond-mat.str-el]. (Visited on 07/02/2026). Pre-published (cit. on pp. 2, 5, 31).
- [KZ23] Liang Kong and Hao Zheng. “Categorical Computation”. In: *Frontiers of Physics* 18.2 (Apr. 2023), p. 21302. doi: [10.1007/s11467-022-1251-5](https://doi.org/10.1007/s11467-022-1251-5). arXiv: [2102.04814](https://arxiv.org/abs/2102.04814) [quant-ph]. (Visited on 07/01/2026) (cit. on p. 4).
- [LDV24] Laurens Lootens, Clement Delcamp, and Frank Verstraete. “Dualities in One-Dimensional Quantum Lattice Models: Topological Sectors”. In: *PRX Quantum* 5.1 (Mar. 6, 2024), p. 010338. doi: [10.1103/PRXQuantum.5.010338](https://doi.org/10.1103/PRXQuantum.5.010338). arXiv: [2211.03777](https://arxiv.org/abs/2211.03777) [quant-ph]. (Visited on 06/08/2026) (cit. on p. 34).
- [Lee26] Seung-Sup Lee. *Ssblee/Telum.jl*. June 30, 2026. URL: <https://github.com/ssblee/Telum.jl> (visited on 07/15/2026) (cit. on p. 41).
- [Liu+22] Yu-Jie Liu et al. “Methods for Simulating String-Net States and Anyons on a Digital Quantum Computer”. In: *PRX Quantum* 3.4 (Nov. 7, 2022), p. 040315. doi: [10.1103/PRXQuantum.3.040315](https://doi.org/10.1103/PRXQuantum.3.040315). (Visited on 07/16/2026) (cit. on p. 4).
- [LM77] J. M. Leinaas and J. Myrheim. “On the Theory of Identical Particles”. In: *Il Nuovo Cimento B (1971-1996)* 37.1 (Jan. 1, 1977), pp. 1–23. doi: [10.1007/BF02727953](https://doi.org/10.1007/BF02727953). (Visited on 07/19/2026) (cit. on pp. 3, 16).
- [McM19] Nathan McMahon. *Qnla/SymLibrary*. Mar. 4, 2019. URL: <https://github.com/qnla/SymLibrary> (visited on 07/05/2026) (cit. on p. 41).
- [Men26] Christian B. Mendl. *Qc-Tum/Chemtensor*. qc-tum, May 28, 2026. URL: <https://github.com/qc-tum/chemtensor> (visited on 06/30/2026) (cit. on p. 41).
- [MG02] I. P. McCulloch and M. Gulácsi. “The Non-Abelian Density Matrix Renormalization Group Algorithm”. In: *Europhysics Letters* 57.6 (Mar. 1, 2002), p. 852. doi: [10.1209/epl/i2002-00393-0](https://doi.org/10.1209/epl/i2002-00393-0). (Visited on 07/16/2026) (cit. on pp. 3, 41).
- [Moc03] Carlos Mochon. “Anyons from Nonsolvable Finite Groups Are Sufficient for Universal Quantum Computation”. In: *Physical Review A* 67.2 (2003). doi: [10.1103/PhysRevA.67.022315](https://doi.org/10.1103/PhysRevA.67.022315) (cit. on p. 4).

- [Moc04] Carlos Mochon. “Anyon Computers with Smaller Groups”. In: *Physical Review A* 69.3 (Mar. 11, 2004), p. 032306. DOI: [10.1103/PhysRevA.69.032306](https://doi.org/10.1103/PhysRevA.69.032306). arXiv: [quant-ph/0306063](https://arxiv.org/abs/quant-ph/0306063). (Visited on 07/01/2026) (cit. on p. 4).
- [Mos25] Milo Moses. “Fusion Categories”. 2025. URL: <https://milomoses.info/fusion-talk.pdf> (visited on 06/16/2026) (cit. on p. 28).
- [MSB20] Nathan A. McMahon, Sukhbinder Singh, and Gavin K. Brennen. “A Holographic Duality from Lifted Tensor Networks”. In: *npj Quantum Information* 6.1 (Apr. 24, 2020), p. 36. DOI: [10.1038/s41534-020-0255-7](https://doi.org/10.1038/s41534-020-0255-7). (Visited on 07/05/2026) (cit. on p. 41).
- [MSS24] David Jaz Myers, Hisham Sati, and Urs Schreiber. “Topological Quantum Gates in Homotopy Type Theory”. In: *Communications in Mathematical Physics* 405.7 (July 2024), p. 172. DOI: [10.1007/s00220-024-05020-8](https://doi.org/10.1007/s00220-024-05020-8). arXiv: [2303.02382](https://arxiv.org/abs/2303.02382) [[quant-ph](#)]. (Visited on 07/17/2026) (cit. on p. 4).
- [Nay+08] Chetan Nayak et al. “Non-Abelian Anyons and Topological Quantum Computation”. In: *Reviews of Modern Physics* 80.3 (Sept. 12, 2008), pp. 1083–1159. DOI: [10.1103/RevModPhys.80.1083](https://doi.org/10.1103/RevModPhys.80.1083). arXiv: [0707.1889](https://arxiv.org/abs/0707.1889) [[cond-mat.str-el](#)]. (Visited on 07/17/2026) (cit. on p. 4).
- [nLa26] nLab authors. *Spherical Braid Group*. July 2026. URL: <https://ncatlab.org/nlab/show/spherical+braid+group> (cit. on p. 18).
- [OR95] Stellan Ostlund and Stefan Rommer. “Thermodynamic Limit of the Density Matrix Renormalization for the Spin-1 Heisenberg Chain”. In: *Physical Review Letters* 75.19 (Nov. 6, 1995), pp. 3537–3540. DOI: [10.1103/PhysRevLett.75.3537](https://doi.org/10.1103/PhysRevLett.75.3537). arXiv: [cond-mat/9503107](https://arxiv.org/abs/cond-mat/9503107). (Visited on 07/15/2026) (cit. on p. 3).
- [Pen71] Roger Penrose. “Applications of Negative Dimensional Tensors”. In: (1971). Ed. by D.J.A. Welsh, pp. 221–224. URL: <https://ncatlab.org/nlab/files/PenroseNegativeDimensionalTensors.pdf> (visited on 06/30/2026) (cit. on pp. 3, 19).
- [Per+08] D. Perez-Garcia et al. “String Order and Symmetries in Quantum Spin Lattices”. In: *Physical Review Letters* 100.16 (Apr. 22, 2008), p. 167202. DOI: [10.1103/PhysRevLett.100.167202](https://doi.org/10.1103/PhysRevLett.100.167202). arXiv: [0802.0447](https://arxiv.org/abs/0802.0447) [[cond-mat.str-el](#)]. (Visited on 06/28/2026) (cit. on p. 15).
- [Pfe+10] Robert N. C. Pfeifer et al. “Simulation of Anyons with Tensor Network Algorithms”. In: *Physical Review B* 82.11 (Sept. 29, 2010), p. 115126. DOI: [10.1103/PhysRevB.82.115126](https://doi.org/10.1103/PhysRevB.82.115126). arXiv: [1006.3532](https://arxiv.org/abs/1006.3532) [[cond-mat.str-el](#)]. (Visited on 06/16/2026) (cit. on pp. 2, 4, 19, 35).
- [Pfe12] Robert N. C. Pfeifer. *Simulation of Anyons Using Symmetric Tensor Network Algorithms*. Feb. 10, 2012. DOI: [10.48550/arXiv.1202.1522](https://doi.org/10.48550/arXiv.1202.1522). arXiv: [1202.1522](https://arxiv.org/abs/1202.1522) [[cond-mat](#)]. (Visited on 05/11/2026). Pre-published (cit. on pp. 4, 19, 35).
- [Pfe14] Robert N. C. Pfeifer. “Measures of Entanglement in Non-Abelian Anyonic Systems”. In: *Physical Review B* 89.3 (Jan. 6, 2014), p. 035105. DOI: [10.1103/PhysRevB.89.035105](https://doi.org/10.1103/PhysRevB.89.035105). (Visited on 07/05/2026) (cit. on p. 4).
- [Ram+25] Marek M. Rams et al. “YASTN: Yet Another Symmetric Tensor Networks; A Python Library for Abelian Symmetric Tensor Network Calculations”. In: *SciPost Phys. Codebases* (2025), p. 52. DOI: [10.21468/SciPostPhysCodeb.52](https://doi.org/10.21468/SciPostPhysCodeb.52) (cit. on p. 41).
- [Rie25] Jeremias Rieser. *jkr11/Exactsynth-Hs*. Nov. 20, 2025. URL: <https://github.com/jkr11/exactsynth-hs> (visited on 08/03/2026) (cit. on p. 38).
- [Rob+19] Chase Roberts et al. *TensorNetwork: A Library for Physics and Machine Learning*. 2019. arXiv: [1905.01330](https://arxiv.org/abs/1905.01330) [[physics.comp-ph](#)] (cit. on p. 41).
- [RSW09] Eric Rowell, Richard Stong, and Zhenghan Wang. *On Classification of Modular Tensor Categories*. Nov. 9, 2009. DOI: [10.48550/arXiv.0712.1377](https://doi.org/10.48550/arXiv.0712.1377). arXiv: [0712.1377](https://arxiv.org/abs/0712.1377) [[math.QA](#)]. (Visited on 09/23/2026). Pre-published (cit. on p. 19).
- [RT91] N. Reshetikhin and V. G. Turaev. “Invariants of 3-Manifolds via Link Polynomials and Quantum Groups”. In: *Inventiones mathematicae* 103.1 (Dec. 1, 1991), pp. 547–597. DOI: [10.1007/BF01239527](https://doi.org/10.1007/BF01239527). (Visited on 07/15/2026) (cit. on pp. 4, 19).

- [Sch+20] Philipp Schmoll et al. “A Programming Guide for Tensor Networks with Global $SU(2)$ Symmetry”. In: *Annals of Physics* 419 (Aug. 1, 2020), p. 168232. doi: [10.1016/j.aop.2020.168232](https://doi.org/10.1016/j.aop.2020.168232). (Visited on 05/07/2026) (cit. on pp. 3, 19).
- [Sch05] Issai Schur. “Neue Begründungen der Theorie der Gruppencharaktere”. In: *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften zu Berlin* (Januar bis Juni 1905), pp. 406–433. URL: http://archive.org/details/bub_gb_KwUoAAAAYAAJ (visited on 07/15/2026) (cit. on p. 25).
- [Sch20] Philipp Schmoll. “Tensor Network Simulations with Global $SU(2)$ Symmetry”. Mainz: Johannes Gutenberg University, July 2020. 150 pp. URL: <https://openscience.ub.uni-mainz.de/server/api/core/bitstreams/d9d9ca4e-46c8-4a8e-a831-7611475d2504/content> (visited on 05/08/2026) (cit. on p. 3).
- [Sel11] P. Selinger. “A Survey of Graphical Languages for Monoidal Categories”. In: *New Structures for Physics*. Ed. by Bob Coecke. Berlin, Heidelberg: Springer, 2011, pp. 289–355. ISBN: 978-3-642-12821-9. doi: [10.1007/978-3-642-12821-9_4](https://doi.org/10.1007/978-3-642-12821-9_4). arXiv: [0908.3347v1](https://arxiv.org/abs/0908.3347v1) [math.CT] (cit. on pp. 4, 19).
- [Sim+06] S. H. Simon et al. “Topological Quantum Computing with Only One Mobile Quasiparticle”. In: *Physical Review Letters* 96.7 (Feb. 23, 2006), p. 070503. doi: [10.1103/PhysRevLett.96.070503](https://doi.org/10.1103/PhysRevLett.96.070503). arXiv: [quant-ph/0509175](https://arxiv.org/abs/quant-ph/0509175). (Visited on 07/19/2026) (cit. on pp. 4, 37).
- [Sim23] Steven H. Simon. *Topological Quantum*. Oxford University Press, Sept. 29, 2023. ISBN: 978-0-19-888672-3. doi: [10.1093/oso/9780198886723.001.0001](https://doi.org/10.1093/oso/9780198886723.001.0001). (Visited on 06/24/2026) (cit. on pp. 4, 18).
- [Sin+14] Sukhwinder Singh et al. “Matrix Product States for Anyonic Systems and Efficient Simulation of Dynamics”. In: *Physical Review B* 89.7 (Feb. 13, 2014), p. 075112. doi: [10.1103/PhysRevB.89.075112](https://doi.org/10.1103/PhysRevB.89.075112). (Visited on 07/08/2026) (cit. on p. 36).
- [SM98] G. Sierra and M. A. Martin-Delgado. *The Density Matrix Renormalization Group, Quantum Groups and Conformal Field Theory*. Version 3. Nov. 30, 1998. doi: [10.48550/arXiv.cond-mat/9811170](https://doi.org/10.48550/arXiv.cond-mat/9811170). arXiv: [cond-mat/9811170](https://arxiv.org/abs/cond-mat/9811170). (Visited on 07/16/2026). Pre-published (cit. on p. 3).
- [SMB18] Sukhwinder Singh, Nathan A. McMahon, and Gavin K. Brennen. “Holographic Spin Networks from Tensor Network States”. In: *Physical Review D* 97.2 (Jan. 26, 2018), p. 026013. doi: [10.1103/PhysRevD.97.026013](https://doi.org/10.1103/PhysRevD.97.026013). arXiv: [1702.00392](https://arxiv.org/abs/1702.00392) [cond-mat.str-el]. (Visited on 05/11/2026) (cit. on p. 3).
- [SO20] Philipp Schmoll and Roman Orus. “Benchmarking Global $SU(2)$ Symmetry in 2d Tensor Network Algorithms”. In: *Physical Review B* 102.24 (Dec. 1, 2020), p. 241101. doi: [10.1103/PhysRevB.102.241101](https://doi.org/10.1103/PhysRevB.102.241101). arXiv: [2005.02748](https://arxiv.org/abs/2005.02748) [cond-mat]. (Visited on 05/07/2026) (cit. on p. 3).
- [SPV10] Sukhwinder Singh, Robert N. C. Pfeifer, and Guifre Vidal. “Tensor Network Decompositions in the Presence of a Global Symmetry”. In: *Physical Review A* 82.5 (Nov. 8, 2010), p. 050301. doi: [10.1103/PhysRevA.82.050301](https://doi.org/10.1103/PhysRevA.82.050301). arXiv: [0907.2994](https://arxiv.org/abs/0907.2994) [cond-mat.str-el]. (Visited on 05/11/2026) (cit. on pp. 3, 14, 41).
- [SPV11] Sukhwinder Singh, Robert N. C. Pfeifer, and Guifre Vidal. “Tensor Network States and Algorithms in the Presence of a Global $U(1)$ Symmetry”. In: *Physical Review B* 83.11 (Mar. 15, 2011), p. 115125. doi: [10.1103/PhysRevB.83.115125](https://doi.org/10.1103/PhysRevB.83.115125). arXiv: [1008.4774v1](https://arxiv.org/abs/1008.4774v1) [cond-mat.str-el]. (Visited on 05/06/2026) (cit. on p. 3).
- [SS22] Hisham Sati and Urs Schreiber. “Differential Cohomotopy Implies Intersecting Brane Observables via Configuration Spaces and Chord Diagrams”. In: *Advances in Theoretical and Mathematical Physics* 26.4 (2022), pp. 957–1051. doi: [10.4310/ATMP.2022.v26.n4.a4](https://doi.org/10.4310/ATMP.2022.v26.n4.a4). arXiv: [1912.10425](https://arxiv.org/abs/1912.10425) [hep-th]. (Visited on 09/08/2026) (cit. on p. 16).
- [SS23] Hisham Sati and Urs Schreiber. “Anyonic Topological Order in Twisted Equivariant Differential (TED) K-Theory”. In: *Reviews in Mathematical Physics* 35.03 (Apr. 2023), p. 2350001. doi: [10.1142/S0129055X23500010](https://doi.org/10.1142/S0129055X23500010). arXiv: [2206.13563](https://arxiv.org/abs/2206.13563) [hep-th]. (Visited on 09/09/2026) (cit. on p. 18).

- [SS26] Hisham Sati and Urs Schreiber. “Entanglement of Sections: The Pushout of Entangled and Parameterized Quantum Information”. In: *Quantum Studies: Mathematics and Foundations* 13.2 (June 2026), p. 23. doi: [10.1007/s40509-026-00397-8](https://doi.org/10.1007/s40509-026-00397-8). arXiv: [2309.07245](https://arxiv.org/abs/2309.07245) [quant-ph]. (Visited on 07/19/2026) (cit. on p. 4).
- [SV12] Sukhwinder Singh and Guifre Vidal. “Tensor Network States and Algorithms in the Presence of a Global SU(2) Symmetry”. In: *Physical Review B* 86.19 (Nov. 9, 2012), p. 195114. doi: [10.1103/PhysRevB.86.195114](https://doi.org/10.1103/PhysRevB.86.195114). arXiv: [1208.3919v1](https://arxiv.org/abs/1208.3919v1) [cond-mat.str-el]. (Visited on 05/11/2026) (cit. on pp. 2, 3, 30).
- [SWD11] Mehdi Saeedi, Robert Wille, and Rolf Drechsler. “Synthesis of Quantum Circuits for Linear Nearest Neighbor Architectures”. In: *Quantum Information Processing* 10.3 (June 2011), pp. 355–377. doi: [10.1007/s11128-010-0201-2](https://doi.org/10.1007/s11128-010-0201-2). arXiv: [1110.6412](https://arxiv.org/abs/1110.6412) [quant-ph]. (Visited on 09/10/2026) (cit. on p. 38).
- [Tre+08] Simon Trebst et al. “A Short Introduction to Fibonacci Anyon Models”. In: *Progress of Theoretical Physics Supplement* 176 (June 1, 2008), pp. 384–407. doi: [10.1143/PTPS.176.384](https://doi.org/10.1143/PTPS.176.384). arXiv: [0902.3275v1](https://arxiv.org/abs/0902.3275v1) [cond-mat.stat-mech]. (Visited on 06/28/2026) (cit. on pp. 4, 35).
- [Unf24] Jakob Unfried. “Advancing Tensor Network Simulations -”. PhD thesis. Technische Universität München, 2024, p. 208. URL: <https://mediatum.ub.tum.de/1755273> (cit. on pp. 2, 3, 19, 35, 41).
- [Val21a] Sachin J. Valera. “Fusion Structure from Exchange Symmetry in (2+1)-Dimensions”. In: *Annals of Physics* 429 (June 2021), p. 168471. doi: [10.1016/j.aop.2021.168471](https://doi.org/10.1016/j.aop.2021.168471). arXiv: [2004.06282](https://arxiv.org/abs/2004.06282) [quant-ph]. (Visited on 09/09/2026) (cit. on p. 18).
- [Val21b] Sachin Jayesh Valera. “Topological Quantum and Skein-Theoretic Aspects of Braided Fusion Categories”. PhD thesis. Norway: University of Bergen, 2021. ISBN: 978-82-308-5158-6 978-82-308-6990-1. URL: [11250/2770245](https://nbn-resolving.org/urn:nbn:no:bi-11250-2770245) (visited on 06/08/2026) (cit. on p. 27).
- [VC04] F. Verstraete and J. I. Cirac. *Renormalization Algorithms for Quantum-Many Body Systems in Two and Higher Dimensions*. July 2, 2004. doi: [10.48550/arXiv.cond-mat/0407066](https://doi.org/10.48550/arXiv.cond-mat/0407066). arXiv: [cond-mat/0407066](https://arxiv.org/abs/cond-mat/0407066). (Visited on 07/15/2026). Pre-published (cit. on p. 3).
- [Vid08] G. Vidal. “A Class of Quantum Many-Body States That Can Be Efficiently Simulated”. In: *Physical Review Letters* 101.11 (Sept. 12, 2008), p. 110501. doi: [10.1103/PhysRevLett.101.110501](https://doi.org/10.1103/PhysRevLett.101.110501). arXiv: [quant-ph/0610099](https://arxiv.org/abs/quant-ph/0610099). (Visited on 07/15/2026) (cit. on p. 3).
- [VW18] Gert Vercleyen, Frank Verstraete, and Universiteit Gent. Faculteit Wetenschappen. “The Mathematical Structure of Tensor Networks”. MA thesis. Master of Science in de wiskunde, 2018. URL: https://lib.ugent.be/fulltxt/RUG01/002/479/664/RUG01-002479664_2018_0001_AC.pdf (cit. on pp. 4, 15).
- [Wei12a] Andreas Weichselbaum. “Non-Abelian Symmetries in Tensor Networks: A Quantum Symmetry Space Approach”. In: *Annals of Physics* 327.12 (Dec. 1, 2012), pp. 2972–3047. doi: [10.1016/j.aop.2012.07.009](https://doi.org/10.1016/j.aop.2012.07.009). (Visited on 05/07/2026) (cit. on pp. 3, 41).
- [Wei12b] Andreas Weichselbaum. “Tensor Networks and the Numerical Renormalization Group”. In: *Physical Review B* 86.24 (Dec. 20, 2012), p. 245124. doi: [10.1103/PhysRevB.86.245124](https://doi.org/10.1103/PhysRevB.86.245124). (Visited on 06/30/2026) (cit. on p. 2).
- [Wei20] Andreas Weichselbaum. “X-Symbols for Non-Abelian Symmetries in Tensor Networks”. In: *Physical Review Research* 2.2 (June 23, 2020), p. 023385. doi: [10.1103/PhysRevResearch.2.023385](https://doi.org/10.1103/PhysRevResearch.2.023385). (Visited on 06/30/2026) (cit. on p. 3).
- [Wei24] Andreas Weichselbaum. “QSpace - An Open-Source Tensor Library for Abelian and Non-Abelian Symmetries”. In: *SciPost Physics Codebases* (Nov. 14, 2024), p. 40. doi: [10.21468/SciPostPhysCodeb.40](https://doi.org/10.21468/SciPostPhysCodeb.40). arXiv: [2405.06632v1](https://arxiv.org/abs/2405.06632v1) [cond-mat.str-el]. (Visited on 05/08/2026) (cit. on p. 41).
- [Whi92] Steven R. White. “Density Matrix Formulation for Quantum Renormalization Groups”. In: *Physical Review Letters* 69.19 (Nov. 9, 1992), pp. 2863–2866. doi: [10.1103/PhysRevLett.69.2863](https://doi.org/10.1103/PhysRevLett.69.2863). (Visited on 07/15/2026) (cit. on p. 3).

- [Whi93] Steven R. White. “Density-Matrix Algorithms for Quantum Renormalization Groups”. In: *Physical Review B* 48.14 (Oct. 1, 1993), pp. 10345–10356. doi: [10.1103/PhysRevB.48.10345](https://doi.org/10.1103/PhysRevB.48.10345). (Visited on 07/15/2026) (cit. on p. 3).
- [Wil90] Frank Wilczek. *Fractional Statistics and Anyon Superconductivity*. WORLD SCIENTIFIC, 1990. doi: [10.1142/0961](https://doi.org/10.1142/0961). eprint: <https://www.worldscientific.com/doi/pdf/10.1142/0961> (cit. on pp. 3, 16).
- [Wou+14] Sebastian Wouters et al. “CheMPS2: A Free Open-Source Spin-Adapted Implementation of the Density Matrix Renormalization Group for Ab Initio Quantum Chemistry”. In: *Computer Physics Communications* 185.6 (June 2014), pp. 1501–1514. doi: [10.1016/j.cpc.2014.01.019](https://doi.org/10.1016/j.cpc.2014.01.019). arXiv: [1312.2415](https://arxiv.org/abs/1312.2415) [cond-mat.str-el]. (Visited on 09/16/2026) (cit. on p. 41).
- [Wu+25] Kai-Hsin Wu et al. “The Cytex Library for Tensor Networks”. In: *SciPost Physics Codebases* (Mar. 13, 2025), p. 53. doi: [10.21468/SciPostPhysCodeb.53](https://doi.org/10.21468/SciPostPhysCodeb.53). arXiv: [2401.01921](https://arxiv.org/abs/2401.01921) [cs.MS]. (Visited on 09/16/2026) (cit. on p. 41).
- [Wu84] Yong-Shi Wu. “General Theory for Quantum Statistics in Two Dimensions”. In: *Physical Review Letters* 52.24 (June 11, 1984), pp. 2103–2106. doi: [10.1103/PhysRevLett.52.2103](https://doi.org/10.1103/PhysRevLett.52.2103). (Visited on 09/08/2026) (cit. on p. 16).
- [Yos26] Atis Yosprakob. *Ayosprakob/Grassmannn*. June 7, 2026. URL: <https://github.com/ayosprakob/grassmannn> (visited on 06/30/2026) (cit. on p. 41).
- [Zha+23] Huanchen Zhai et al. “Block2: A Comprehensive Open Source Framework to Develop and Apply State-of-the-Art DMRG Algorithms in Electronic Structure and Beyond”. In: *The Journal of Chemical Physics* 159.23 (Dec. 18, 2023), p. 234801. doi: [10.1063/5.0180424](https://doi.org/10.1063/5.0180424). (Visited on 09/17/2026) (cit. on p. 41).